CYBERSECURITY LABORATORY SERIES

Building a Virtual Machine Laboratory

VMware Workstation, VirtualBox, VMware Fusion, and UTM

A Hands-On Lab Guide for Students and Instructors

Windows • Linux (Fedora) • macOS

Joshua Moses

CyberTrace Foundation

August 2026 Edition

How to Use This Guide

This guide does two jobs. Part 1 explains what virtualization actually does to a computer, which is the part students need in order to make good decisions later. Parts 2 through 5 are step-by-step procedures for four hypervisors across three host operating systems. Part 6 is the reference material you reach for when something does not work.

Every major procedure follows the same three-beat structure:

Beat	What it does	Why it is there
Explain	States what the procedure accomplishes and which decisions matter	A student who knows why a step exists can recover when the interface changes
Perform	Numbered steps with exact commands, exact menu paths, and exact values	Reproducible by a student working alone at 11pm
Verify	A specific test that proves the step worked	Catches silent failures before they compound into a broken lab

Procedures are followed by a **Troubleshooting** table covering the failures that actually happen, rather than every failure that theoretically could.

Conventions

Element	Meaning
monospace text	A command, filename, path, or exact value to type
Bold	A user interface element to click, or an emphasized decision
Note	Background that helps but is not required to complete the step
Tip	A shortcut or a practice worth adopting
Warning	Something that will cost you time, data, or security if ignored
Verify	The checkpoint that proves the procedure worked
[SCREENSHOT REQUIRED]	A placeholder for the instructor to capture a current image, with capture instructions

Prerequisites

This guide assumes a student can install ordinary applications, navigate a file system, and open a terminal or command prompt. It does not assume prior virtualization experience, and every term is explained before it is used. Appendix E collects the definitions in one place.

Scope and Currency

Software versions are recorded in Appendix F. All four products move, and the Broadcom download path in particular has changed more than once since the VMware acquisition. The procedures here are stable; the version numbers and the exact wording of a menu item are not. Before teaching from this guide, spend ten minutes confirming that the download paths still resolve.

> **WARNING**
> Everything in this guide is written for lab environments you own or are explicitly authorized to use. Evaluation licences carry real terms, the VirtualBox Extension Pack is free only for personal and educational use, and network isolation in a security lab is a genuine safety control rather than a formality. Read the licence screens instead of clicking past them, and teach students to do the same.

Contents

Part 1: How Virtualization Works

Everything in Part 2 onward is button clicking. This part is the reason the buttons matter. A student who installs a hypervisor without understanding what it does to the host will build a lab that thrashes, crashes, or quietly corrupts itself, and will not know why. Twenty minutes here saves a semester of mystery.

1. How Virtual Machines Use the Processor, Memory, and Disk of the Host Computer

A virtual machine is not a program that pretends to be a computer. It is a set of real hardware resources that the host lends out under supervision, plus a small amount of software theater to make the guest operating system believe it owns them. That distinction drives every performance decision you will make in this course.

1.1 The Hypervisor

The software that does the lending is called a **hypervisor**. It sits between physical hardware and the guest operating systems, and it decides which guest gets which resource, and for how long.

There are two types, and the difference matters for what you are about to install.

	Type 1 (bare metal)	Type 2 (hosted)
Where it runs	Directly on the hardware. It **is** the operating system.	As an application on top of Windows, Linux, or macOS.
Examples	VMware ESXi, Microsoft Hyper-V (server role), Proxmox VE, KVM	VMware Workstation, VMware Fusion, Oracle VirtualBox, UTM, Parallels Desktop
Overhead	Very low. Nothing competes with it.	Higher. The host OS, your browser, and Teams are all still running.
Typical use	Data center, production, dedicated lab server	Laptop and workstation labs, training, malware analysis, development
This guide	Out of scope	**All four products covered here are Type 2**

Everything you install in this course is a Type 2 hypervisor. That means the host operating system never goes away. It keeps its own memory, keeps scheduling its own threads, and keeps writing to the same disk your virtual machines live on. Every resource conversation from here forward is a conversation about sharing.

> **NOTE**
> Modern Type 2 hypervisors are not emulators in the classic sense. They do not interpret guest instructions one at a time. They hand guest instructions directly to the physical CPU and only intervene when the guest tries to do something privileged. That is why a virtual machine feels close to native speed, and it is also why hardware virtualization support has to be turned on before any of this works.

1.2 The Processor

Hardware virtualization extensions

Intel calls it **VT-x**. AMD calls it **AMD-V**. Both do the same job: they add a new, more privileged execution mode below the operating system so a hypervisor can run there and let guest code execute natively on the silicon. Without these extensions, virtualization falls back to slow software emulation, and most modern hypervisors simply refuse to start a 64-bit guest at all.

A second extension matters almost as much. Intel calls it **EPT** (Extended Page Tables), AMD calls it **RVI** or **NPT** (Nested Page Tables). It lets the CPU translate guest memory addresses to physical memory addresses in hardware. Before this existed, the hypervisor maintained shadow page tables in software and memory-heavy workloads crawled.

> **WARNING**
> These features are frequently disabled in UEFI/BIOS from the factory, and some corporate images disable them deliberately. If a virtual machine will not start, or starts in "unaccelerated" mode, check the firmware first, before you troubleshoot anything else. On Windows hosts, an active Hyper-V, Windows Sandbox, Credential Guard, or Memory Integrity feature can also claim the extensions and leave nothing for other hypervisors. Section 22 covers that conflict in detail.

Virtual CPUs are time slices, not chips

When you assign a virtual machine 4 vCPUs, you have not given it four processors. You have given it four schedulable threads that the hypervisor will place onto physical cores when time is available. The guest kernel sees four CPUs and schedules across them. The hypervisor sees four more things in a queue, alongside the host operating system, your browser, your endpoint agent, and every other running VM.

Two consequences follow, and students get both of them wrong at first.

- **Total assigned vCPUs may exceed physical cores.** This is normal and expected. A host with 8 cores can comfortably run five VMs with 2 vCPUs each, as long as they are not all busy at the same moment. This is called oversubscription.
- **Assigning more vCPUs to one VM often makes it slower.** Hypervisors generally try to schedule a multi-vCPU guest so its virtual CPUs run at roughly the same time. The more vCPUs a VM has, the longer it waits for that many cores to be free at once. A 2 vCPU domain controller will usually outperform an 8 vCPU domain controller on a 4-core laptop.

> **TIP**
> Start every lab virtual machine at 2 vCPUs. Raise it only when you have measured a CPU bottleneck inside the guest. "More is better" is the single most common sizing mistake in student labs.

Cores, threads, and what to count

A CPU advertised as "8 cores, 16 threads" has eight physical cores with simultaneous multithreading (Intel Hyper-Threading or AMD SMT). Hypervisors will let you allocate against the 16 logical processors, but a second thread on a busy core delivers perhaps twenty to thirty percent of a real core, not a hundred. Size your lab against physical cores and treat the extra threads as headroom.

On Apple Silicon there is a further wrinkle. An M-series chip has performance cores and efficiency cores. Virtualization frameworks generally place guest vCPUs on performance cores, so an M2 with four performance cores has less practical capacity for concurrent VMs than the marketing core count suggests.

1.3 Memory

Memory is where lab plans die. CPU contention makes a virtual machine slow. Memory contention makes the entire host unusable, including the tools you were going to use to fix it.

How guest RAM is provided

When a virtual machine is assigned 4 GB of RAM, the hypervisor creates a 4 GB region the guest believes is physical memory. The guest kernel builds its page tables against that region. The CPU translates guest-physical addresses to host-physical addresses in hardware, using EPT or NPT.

Where those pages actually live is the hypervisor's decision. They may sit in host RAM, or they may be pushed out to a swap file on the host disk. To the guest, both look identical. The only difference is that one is roughly a hundred nanoseconds away and the other is roughly a hundred microseconds away, a difference of three orders of magnitude.

Reserved versus dynamic allocation

Model	What happens	Where you see it
Fully reserved	The host commits the full amount when the VM powers on. Predictable, and it fails fast if the memory is not there.	VMware "Fit all virtual machine memory into reserved host RAM"
Partly reserved (default)	Some of the guest memory may be swapped to disk under pressure. Good density, occasional stalls.	VMware Workstation and Fusion default behavior
Ballooning	A driver inside the guest inflates a balloon, forcing the guest to release pages it considers free, which the host reclaims.	VMware Tools, VirtualBox Guest Additions
Host swapping	The host pages guest memory to its own swap file. This is the slow path and it is what "everything froze" usually means.	All Type 2 hypervisors under pressure

The host reserve rule

The host operating system is not free. A modern Windows 11 host with an EDR agent, a browser, and a mail client will use 6 to 8 GB before you start a single virtual machine. macOS behaves similarly. Fedora with a GNOME desktop is lighter but still wants 2 to 3 GB.

Use this rule when planning, and treat it as a floor rather than a target:

```
Usable VM memory = Installed RAM - Host reserve

  Windows 11 host with security tooling ....... reserve 8 GB
  macOS host ................................. reserve 8 GB
```

```
Fedora Workstation host ..................... reserve 4 GB
Fedora Server / minimal Linux host .......... reserve 2 GB
```

A 16 GB Windows laptop therefore has about 8 GB to spend across all running virtual machines. That is one Windows Server VM at 4 GB plus one Linux VM at 2 GB, with a little left over. It is not three Windows Servers, no matter what the sum on paper looks like.

> **WARNING**
> Never assign a single virtual machine more memory than the host has free. Some hypervisors will let you do it. The result is continuous host swapping, and a laptop with a spinning disk in that state can take twenty minutes to become responsive again.

1.4 Disk Space

A virtual disk is a file

The virtual hard disk your guest formats and installs onto is one or more ordinary files on the host filesystem. The formats differ by product:

Format	Extension	Used by
VMware virtual disk	.vmdk	VMware Workstation, VMware Fusion, ESXi
VirtualBox disk image	.vdi	Oracle VirtualBox (native default)
Virtual hard disk	.vhd / .vhdx	Microsoft Hyper-V, readable by VirtualBox
QEMU copy-on-write	.qcow2	QEMU/KVM, UTM (QEMU backend)
Raw image	.img / .raw	UTM (Apple Virtualization backend), many tools

When the guest writes to what it thinks is sector 1,048,576 of its disk, the hypervisor translates that into an offset inside the file and asks the host filesystem to write there. Every guest disk operation is therefore a host file operation, subject to the host filesystem, the host cache, the host encryption layer, and the host antivirus scanner.

Thin versus thick provisioning

	Thin (allocate as needed)	Thick (allocate now)
Initial size on disk	A few hundred megabytes	The full declared size
Growth	Grows as the guest writes	Fixed at creation
Write performance	Slightly slower, the file must extend	Slightly faster, blocks are already there
Fragmentation	Higher over time	Lower
Risk	Host can run out of space while the guest still believes it has room	None of that, but you pay up front
Recommended for	Almost all lab use	Performance testing, or a lab on a dedicated data drive

Thin provisioning is the default nearly everywhere and it is the right default for a teaching lab. Declare a 100 GB virtual disk for a Windows Server VM and it will occupy roughly 15 GB after installation. Just remember what you have promised. Six thin 100 GB disks on a 512 GB SSD is a promise of 600 GB you cannot keep.

> **WARNING**
> When a host filesystem fills up while virtual machines are running, the guests do not see a graceful disk-full error. They see failed writes at unpredictable moments. Database corruption and unbootable guests are the normal outcome. Keep at least 15 percent of the host volume free at all times.

Snapshots and why they get expensive

A snapshot freezes the current disk state and redirects all subsequent writes into a new delta file. The original file becomes read-only. This is the single most valuable feature in a security lab, because it turns "I destroyed the domain controller" into a thirty-second recovery.

The cost is storage and speed. Each snapshot adds a delta file that grows with every write. A chain of ten snapshots means a read may have to walk ten files to find the current version of a block. Two or three snapshots per lab VM is healthy. Twenty is a self-inflicted performance problem.

> **TIP**
> Take a snapshot named Clean Build immediately after you install the guest operating system, install guest tools, and apply updates. That one snapshot will save you more time than every other snapshot you ever take.

Storage type matters more than storage size

An NVMe SSD delivers hundreds of thousands of IOPS. A SATA SSD delivers tens of thousands. A 5400 RPM laptop hard drive delivers roughly 80. Three virtual machines booting at once on a mechanical drive will produce a random I/O pattern that a spinning disk simply cannot serve, and the lab will appear frozen for minutes at a time.

If you are choosing where to spend money on a lab machine, the order is: SSD first, RAM second, CPU third. That order surprises people and it is correct.

1.5 A Note on Network and Devices

The same borrowing model applies to everything else. A virtual NIC is a software device backed by a host adapter through NAT, bridging, or a host-only switch. A virtual GPU is a paravirtualized display driver with no real 3D acceleration unless you go out of your way. USB devices are captured from the host and passed through, which means the host loses them while the guest holds them.

Networking mode selection has direct security consequences in a lab that handles malware or unknown software, so it gets its own treatment in Section 23.

2. Why Plenty of Memory and Disk Space Change Everything

Students often ask whether they can "get by" with 8 GB of RAM and a nearly full 256 GB SSD. The honest answer is that they can complete perhaps a third of a normal security curriculum, slowly, and will spend a meaningful part of the course troubleshooting resource exhaustion instead of learning the subject. Here is what the extra capacity actually buys.

2.1 What Ample Memory Buys You

Concurrency, which is the whole point

A single virtual machine teaches installation. Multiple simultaneous virtual machines teach everything else. Active Directory needs a domain controller and at least one member workstation. A network exercise needs an attacker host, a target, and ideally something in between. An incident response exercise needs a victim and a collector.

Concurrency is bounded almost entirely by RAM. This is what the difference looks like in practice:

Host RAM	Realistic simultaneous lab	What you can teach
8 GB	One Linux VM at 2 GB, or one Windows client at 4 GB, nothing else running	Single-host tooling, command line, basic forensics
16 GB	One Windows Server at 4 GB plus one Linux VM at 2 GB	Basic Active Directory, one attacker and one target
32 GB	Domain controller, member server, Windows client, Kali or Ubuntu attacker, pfSense or similar	Full AD lab, segmented networks, lateral movement, logging pipeline
64 GB and up	All of the above plus SIEM, a detonation VM, and spare capacity	Enterprise-shaped labs, EDR evaluation, malware analysis with isolation

No swapping, which is the difference between slow and unusable

When the host has enough RAM, guest memory stays in RAM. When it does not, the host pages guest memory to disk. The performance cliff is not gradual. A VM whose working set is being swapped does not run at eighty percent speed. It runs at perhaps two percent, and it drags the host down with it because the host is now spending its own I/O bandwidth on paging.

Room for snapshots that hold memory state

A snapshot taken while a virtual machine is running captures RAM as well as disk, so you can resume exactly where you were, mid-attack, mid-debug, mid-capture. That memory image is written to the host and it is the size of the guest's assigned RAM. Ten running snapshots of a 4 GB VM is 40 GB of host disk and it also means the host had to be able to hold that memory in the first place.

The host stays usable

This one is underrated. Instructors run virtual machines alongside a browser, a terminal, a note-taking app, a screen recorder, and a video call. Ample memory means you can demonstrate a lab and talk about it at the same time. Marginal memory means every demonstration is a gamble.

2.2 What Ample Disk Space Buys You

A library instead of a queue

With generous storage you keep a shelf of ready virtual machines: a domain controller, two client builds, three Linux distributions, a firewall appliance, a forensics workstation. With tight storage you delete last week's lab to make room for this week's, which means every lab starts from a fresh install, which costs forty minutes before any teaching begins.

Snapshots and the freedom to break things

Security education requires students to break things on purpose: disable protections, run untrusted binaries, misconfigure permissions, and watch what happens. That is only safe and only practical when reverting is cheap. Snapshots cost disk. Disk therefore buys pedagogical freedom directly.

Realistic data sets

A forensics exercise on a 20 GB disk image, a log analysis exercise on a month of Windows event logs, a packet capture from a busy segment: all of these are storage-hungry, and all of them are what the work actually looks like outside a classroom.

Room for ISO images

The installation media themselves add up quickly, and you want to keep them rather than redownload:

Image	Approximate size
Windows Server 2025 evaluation ISO	5.5 GB
Windows 11 ISO	5.5 GB to 7 GB
Ubuntu Server 26.04 LTS ISO	3 GB
Fedora Server 44 DVD ISO	2.5 GB
Fedora Workstation 44 Live ISO	2.5 GB
A working set of five to eight images	**25 GB to 40 GB before a single VM exists**

2.3 Practical Sizing Targets

These are the numbers to give students who ask what to buy, or what to check before enrolling.

Component	Bare minimum	Comfortable	Recommended for a full security lab
RAM	8 GB	16 GB	**32 GB or more**
Storage type	SATA SSD	NVMe SSD	NVMe SSD, ideally a second dedicated drive
Free storage	100 GB	250 GB	**500 GB to 1 TB**
CPU	4 cores with VT-x or AMD-V	6 to 8 cores	8 or more cores

Per-virtual-machine starting points

Guest	vCPU	RAM	Virtual disk (thin)
Windows Server 2025, Desktop Experience	2	4 GB	80 GB
Windows Server 2025, Server Core	2	2 GB	60 GB
Windows 11 client	2	4 GB	80 GB
Ubuntu Server 26.04 LTS	2	2 GB	40 GB
Fedora Server 44	2	2 GB	40 GB
Fedora Workstation 44 (GUI)	2	4 GB	60 GB
Lightweight Linux tooling VM	2	2 GB	30 GB

> **NOTE**
> These are starting points, not ceilings. Raise memory when the guest is swapping. Raise vCPU only after you have watched CPU saturation inside the guest with Task Manager, top, or htop. Guessing upward wastes capacity that another virtual machine needed.

3. ISO Images and Where to Get Them

3.1 What an ISO Image Actually Is

An ISO image is a byte-for-byte archive of an optical disc, stored in a single file. The name comes from ISO 9660, the standard that defines the filesystem layout on a CD. Modern installation images usually carry a UDF filesystem instead, and often a hybrid layout that also works when written to a USB stick, but the .iso extension stuck.

Three properties are worth understanding:

- **It is a filesystem image, not an archive.** A .zip is a container of compressed files. An ISO is a picture of a disc, including the boot sector, the partition structures, and the directory tree exactly as an optical drive would present them.
- **It is bootable when it carries a boot record.** Installation ISOs include an El Torito boot record for legacy BIOS, an EFI boot image for UEFI, or both. That is what makes them usable as installation media rather than just readable archives.
- **It is read-only by nature.** Nothing writes back into a mounted ISO. This is why installers copy files out of the image onto a target disk.

Why virtualization loves ISO files

Every hypervisor in this guide can attach an ISO file to a virtual machine as a virtual optical drive. No burning, no USB stick, no physical media. The virtual machine boots from it exactly as a physical machine would boot from a DVD. Detaching it later is a menu selection.

This has a practical consequence for your workflow. Keep one ISO library folder on the host, outside any individual virtual machine folder, and point every VM at it. One copy of Windows Server 2025 serves every Windows VM you ever build.

```
Windows host:  C:\ISOs\
Linux host:    /srv/isos/     or   ~/ISOs/
macOS host:    ~/ISOs/
```

WARNING

Do not store the ISO library inside a folder that syncs to OneDrive, iCloud Drive, Google Drive, or Dropbox. Multi-gigabyte files in a sync folder will saturate your uplink, may be silently converted to online-only placeholders, and can make a virtual machine fail to boot when the file is not actually present on disk. This applies with equal force to the folder holding your virtual machines.

3.2 Verifying an ISO Before You Use It

This step is skipped by almost everyone and it should not be, particularly in a security course where we are teaching the habit as much as the task. A truncated download produces an installer that fails halfway through. A tampered image produces something worse.

On Windows (PowerShell)

```
Get-FileHash -Algorithm SHA256 "C:\ISOs\ubuntu-26.04-live-server-amd64.iso"
```

On Linux

```
sha256sum ~/ISOs/ubuntu-26.04-live-server-amd64.iso

# Or verify a whole published checksum file at once:
sha256sum -c SHA256SUMS --ignore-missing
```

On macOS

```
shasum -a 256 ~/ISOs/ubuntu-26.04-live-server-amd64.iso
```

Compare the result against the checksum published on the vendor's download page. Ubuntu and Fedora publish signed checksum files, so you can go one step further and verify the signature on the checksum file itself with GPG. Fedora also publishes a small helper package, and Ubuntu documents the full GPG procedure on its download pages.

TIP

Teach this as a two-minute ritual, not a special occasion. The habit of verifying a downloaded artifact before executing it is the same habit that stops a supply chain compromise from landing.

3.3 Architecture: The Check That Saves an Hour

Before downloading anything, know which instruction set your host uses, because a hypervisor on Apple Silicon cannot boot an x86-64 guest at practical speed.

Host	Architecture	Download this
Intel or AMD PC (Windows or Linux)	x86-64	amd64 / x86_64 / x64 images
Intel Mac	x86-64	amd64 / x86_64 / x64 images
Apple Silicon Mac (M1 through M5)	ARM64	**`arm64` / `aarch64` images**

> **WARNING**
> On an Apple Silicon Mac you need ARM64 installation media. UTM can emulate x86-64, but emulation runs roughly five to twenty times slower than virtualization, and a Windows Server install that takes fifteen minutes natively can take several hours emulated. Sections 18 through 21 handle this in detail, including where the ARM64 Windows media comes from.

3.4 Where to Get Windows Server Evaluation Images

Microsoft publishes fully functional evaluation builds of Windows Server through the Microsoft Evaluation Center. They are free, they include every role and feature, and they are the correct source for classroom and lab use.

The source

```
https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2025

Evaluation Center index:
https://www.microsoft.com/en-us/evalcenter
```

What you get

Property	Detail
Evaluation period	**180 days from installation**
Activation requirement	Must activate over the internet within the first 10 days, otherwise the server begins shutting down automatically
Editions in the ISO	Standard and Datacenter, both in Desktop Experience and Server Core. The installer asks you to choose.
Architecture	x86-64 only. Microsoft has not published an ARM64 Windows Server evaluation ISO.
Approximate size	5.5 GB for the multi-edition ISO
Registration	A Microsoft account and a short profile form
Alternative format	A prebuilt VHD/VHDX is also offered, which is useful for Hyper-V but less so for the products in this guide

Procedure

1. Open the Windows Server 2025 evaluation page listed above.
2. Locate the "Get started for free" section and choose the **ISO** download rather than VHD or Azure.
3. Sign in with a Microsoft account and complete the registration form. Use real information: the country field feeds an export compliance check.
4. Select your language and start the download.
5. Move the finished file into your ISO library folder and verify it against the published hash.

VERIFY

The file is roughly 5.5 GB with a .iso extension. On Windows, right-click the file, choose Properties, and confirm the digital signature is present. A file of a few hundred megabytes is a failed download, not a small edition.

NOTE

Older releases remain available at the Evaluation Center if your curriculum targets them: Windows Server 2022, 2019, and 2016 all still have evaluation listings. Windows Server 2022 is a reasonable choice if you want a slightly more forgiving hardware baseline. If you need a Windows **client** instead of a server, Microsoft publishes Windows 11 ISOs on the Windows download page, and the Windows Insider program publishes ARM64 client images.

WARNING

The 180-day clock starts at installation, not download. Build your gold image, apply updates, install guest tools, then take a snapshot named Clean Build. When the evaluation expires you revert or rebuild rather than losing a semester of work. An evaluation edition can be converted to a licensed edition later with DISM and a valid product key, which is a useful exercise in itself.

3.5 Where to Get Linux Server Distributions

Ubuntu Server

Ubuntu is the most widely deployed Linux server distribution and the one your students are most likely to meet in a job. Long Term Support releases arrive every two years in April and carry five years of standard security maintenance.

```
Primary download page:
https://ubuntu.com/download/server

All releases, torrents, network installer, older versions:
https://releases.ubuntu.com/
https://ubuntu.com/download/alternative-downloads

ARM64 and other architectures (needed for Apple Silicon):
https://cdimage.ubuntu.com/releases/
```

Property	Detail
Current LTS	**Ubuntu Server 26.04 LTS, codename Resolute Raccoon**, released 23 April 2026

Property	Detail
Support window	Five years of standard security maintenance, extendable with Ubuntu Pro
Previous LTS releases	24.04 LTS and 22.04 LTS are still supported and still downloadable
Installer	Subiquity, a text-mode guided installer. There is no graphical desktop in Ubuntu Server.
Architectures	amd64 and arm64 both published, which makes Ubuntu the easiest Linux guest on Apple Silicon
Approximate size	About 3 GB for the live server ISO
Registration	None. Direct download.
Filename pattern	ubuntu-26.04-live-server-amd64.iso / ubuntu-26.04-live-server-arm64.iso

Fedora Server

Fedora sits upstream of Red Hat Enterprise Linux. It moves fast, it ships current kernels and toolchains, and it uses SELinux, firewalld, DNF, and RPM, which makes it the right teaching platform for anyone heading toward RHEL, CentOS Stream, Rocky, or Alma.

```
Fedora Server:
https://fedoraproject.org/server/download

All editions and spins:
https://fedoraproject.org/

Archived releases:
https://dl.fedoraproject.org/pub/fedora/linux/releases/
```

Property	Detail
Current release	**Fedora Linux 44**, released 28 April 2026
Release cadence	Roughly every six months, in April and October
Support window	**About 13 months per release.** Plan to upgrade, or accept that a lab image goes end of life within the year.
Installer	Anaconda, graphical by default even on Server
Editions worth knowing	Server (DVD ISO, plus a small netinstall), Workstation (GNOME live image), IoT, CoreOS
Architectures	x86_64 and aarch64 both published
Approximate size	About 2.5 GB for the Server DVD, under 1 GB for netinstall
Registration	None. Direct download.
Filename pattern	Fedora-Server-dvd-x86_64-44-1.x.iso

> **NOTE**
> Fedora Server versus Fedora Workstation matters for this guide. Sections 6 and 11 install hypervisors on Fedora, and those instructions assume **Fedora Workstation**, because a Type 2 hypervisor with a graphical console needs a desktop session. Fedora Server is what you install as a guest. Do not install VMware Workstation on a headless Fedora Server and expect a usable console.

Choosing between them for teaching

Consideration	Ubuntu Server	Fedora Server
Package manager	apt, .deb	dnf, .rpm
Security framework	AppArmor	**SELinux**, enforcing by default
Firewall front end	ufw (and nftables underneath)	firewalld
Lifecycle	Five years per LTS	About 13 months
Career alignment	Cloud, containers, general Linux estates	Red Hat estates, RHCSA and RHCE paths
Best for a course that	Wants a stable image reused across semesters	Wants students fluent in SELinux, firewalld, and RPM

> **TIP**
> Build one of each. The differences between AppArmor and SELinux, and between ufw and firewalld, are exactly the kind of thing students remember when they have hit both in the same afternoon.

A note on other distributions

Debian, Rocky Linux, AlmaLinux, and openSUSE all publish free ISOs on the same model and all work identically in the four hypervisors covered here. Kali Linux does too, but think twice before making it a student's default Linux VM: it is a specialist penetration testing distribution, it makes a poor general-purpose workstation, and its tooling is precisely what most lab policies prohibit. Teach general Linux on Ubuntu or Fedora, and introduce Kali as a deliberate, scoped tool later.

Part 2: VMware Workstation Pro

VMware Workstation Pro runs on Windows and Linux hosts. Since November 2024 it has been free for commercial, educational, and personal use with no license key required, which makes it a straightforward choice for a classroom. The current release line uses calendar versioning: **26H1**, published in May 2026, replacing the older 17.x numbering.

Two things about this product surprise people who last used it before the Broadcom acquisition. First, the download no longer lives on vmware.com; it lives behind a free account on the Broadcom Support Portal. Second, Workstation Player is no longer a separate cut-down product line in the way it once was. Everything below assumes Workstation Pro.

> **NOTE**
> Because these are free products, they do not come with Broadcom enterprise support. Basic portal accounts cannot open support cases or use live chat. Documentation, the knowledge base, and the Broadcom Communities forums are the support path.

4. Downloading VMware Workstation from the Broadcom Website

4.1 Explain

Broadcom acquired VMware in November 2023 and consolidated distribution into the Broadcom Support Portal. There is no anonymous direct download link any more. You need a free Broadcom account, and you need a completed profile, because the download is gated by an export compliance screening rather than by payment.

This trips up entire classrooms on day one, so build the account before the lab session rather than during it.

4.2 Prerequisites

- An email address you can receive mail at. A personal address (gmail.com, outlook.com) is accepted; a corporate address is not required.
- Roughly 850 MB of free space for the installer.
- A host that meets the requirements: a 64-bit x86-64 CPU with VT-x or AMD-V, and Windows 10 version 21H2 or later, Windows 11, or a supported Linux distribution.

> **WARNING**
> VMware Workstation does not run on Apple Silicon. If your host is an M-series Mac, skip to Part 4 (VMware Fusion) or Part 5 (UTM). VMware Fusion is the Broadcom product for macOS.

4.3 Perform

1. Open a browser and go to the Broadcom home page:

```
https://www.broadcom.com
```

2. In the top right of the page, select **Support Portal**.
3. If you already have an account, choose **Go To Portal** and sign in. If you do not, choose **Register** and create a free Basic account. The direct registration form is:

   ```
   https://profile.broadcom.com/web/registration
   ```

4. Complete your profile fully and accurately, including name, email, street address, and country.

   This is not a marketing form. Broadcom runs an export compliance screening against it, and placeholder data will fail screening and delay or block your download.

5. Once signed in, go to the support portal itself if you were not redirected there automatically:

   ```
   https://support.broadcom.com
   ```

6. Use the division selector near the top of the page and choose **VMware Cloud Foundation**. Workstation and Fusion live under this division, not under end user computing.
7. In the left navigation, select **My Downloads**, then follow the **Free Software Downloads available HERE** link.
8. Search for VMware Workstation Pro and open the product entry.
9. Expand the entry for the current release. As of this writing that is **VMware Workstation Pro 26H1**, and you will see separate entries for Windows and for Linux.
10. Select the platform you need:

    For a Windows host, choose the .exe installer.

    For a Linux host, choose the .bundle installer.

11. The download button is greyed out until you accept the terms. Click the **Terms and Conditions** hyperlink first, which opens the End User Agreement in a new tab. Return to the download tab, and the "I agree to the Terms and Conditions" checkbox will now be selectable. Tick it.
12. Click the download icon. If a Trade Compliance form appears, complete it. Approval is usually immediate but can take longer if your profile is incomplete.

[SCREENSHOT REQUIRED]

Capture: The Broadcom Support Portal My Downloads page with VMware Workstation Pro expanded

Show: The division selector set to VMware Cloud Foundation, the 26H1 release row, and both the Windows and Linux entries visible

Caption: Figure 4-1: Locating the free VMware Workstation Pro download on the Broadcom Support Portal

4.4 Verify

VERIFY

You have a file named similarly to VMware-workstation-full-26H1-<build>.exe (Windows) or VMware-Workstation-Full-26H1-<build>.x86_64.bundle (Linux).

The file is several hundred megabytes. A file of a few kilobytes is an error page saved with the wrong

extension.

4.5 Troubleshooting

Symptom	Cause	Fix
"Complete your profile to gain access"	The profile or trade compliance form is incomplete	Return to your Broadcom profile, fill in every field including street address and country, and wait for screening
The agree checkbox will not tick	You have not opened the Terms and Conditions link	Click the hyperlink itself, let the agreement open, then return to the tab
No Workstation entry in My Downloads	Wrong division selected	Change the division selector to VMware Cloud Foundation
Download fails partway repeatedly	Session timeout on the portal	Sign in again and restart the download; portal sessions are short

5. Installing VMware Workstation on Windows

5.1 Explain

The Windows installer places the hypervisor kernel drivers, the virtual network adapters (VMnet1 for host-only and VMnet8 for NAT), and the Workstation user interface. As of the 26H1 release, Workstation Pro for Windows is a pure 64-bit application throughout, including its installer components and services.

The most important part of this section is not the installer. It is the conflict check in step 1.

5.2 Prerequisites and the Hyper-V Conflict

Windows has a built-in Type 1 hypervisor. When any Windows feature that depends on it is enabled, Windows boots the Hyper-V hypervisor first and everything else runs on top of it. VMware Workstation can operate in that mode, but performance drops noticeably and some features behave differently.

These Windows features all activate the Hyper-V platform, and any one of them is enough:

- Hyper-V role or Hyper-V Platform feature
- Windows Sandbox
- Windows Subsystem for Linux 2 (WSL2)
- Virtual Machine Platform
- Memory Integrity (Core Isolation)
- Credential Guard / Device Guard, frequently pushed by enterprise policy
- Windows Defender Application Guard

Check the current state

Open Task Manager, go to the Performance tab, select CPU, and look at the bottom right. If **Virtualization: Enabled** appears, the CPU extensions are on. If a line reading "A hypervisor has been detected" appears in System Information, Hyper-V is already running.

From an elevated PowerShell prompt:

```
# Is the Hyper-V hypervisor scheduled to launch at boot?
bcdedit /enum {current} | Select-String hypervisorlaunchtype

# Which virtualization-based security features are running?
Get-CimInstance -ClassName Win32_DeviceGuard `
  -Namespace root\Microsoft\Windows\DeviceGuard |
  Select-Object -ExpandProperty SecurityServicesRunning
```

Disable them if you want maximum performance

> **WARNING**
> Disabling these features reduces the security posture of the host. On a personal lab machine this is a reasonable trade. On a managed corporate machine it may violate policy, and Credential Guard may be enforced by Group Policy so that local changes revert at the next policy refresh. Ask before you change it, and prefer a dedicated lab machine.

```
# Run as Administrator. Reboot required afterward.

bcdedit /set hypervisorlaunchtype off

Disable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V-All
-NoRestart
Disable-WindowsOptionalFeature -Online -FeatureName VirtualMachinePlatform
-NoRestart
Disable-WindowsOptionalFeature -Online -FeatureName Containers-
DisposableClientVM -NoRestart

# Then reboot:
Restart-Computer
```

Memory Integrity is switched off separately, under Windows Security > Device security > Core isolation details.

> **TIP**
> To put it back later: bcdedit /set hypervisorlaunchtype auto, re-enable the features, and reboot. Keep the command handy; students who need WSL2 for another class will need it.

5.3 Perform

1. Close every open application. The installer will want to reboot at the end.
2. Right-click the downloaded .exe and choose **Run as administrator**.
3. On the welcome screen, click **Next**.
4. Accept the End User License Agreement and click **Next**.
5. On the Custom Setup screen, review the installation path. The default is fine for most labs. Two checkboxes appear here:

   Add VMware Workstation console tools to system PATH. Tick this. It gives you vmrun and vmware-vdiskmanager at any command prompt, which you will want for automation later.

   Enhanced Keyboard Driver. Optional. Useful if you will send unusual key combinations into guests, or use a non-US keyboard layout in a guest.
6. On the User Experience screen, decide whether to allow startup update checks and the Customer Experience Improvement Program. In a locked-down lab, clear both.
7. Choose whether to create Desktop and Start Menu shortcuts, then click **Next**.
8. Click **Install**. This takes two to five minutes. The screen may flicker as the virtual network adapters are created.
9. Click **Finish**. When prompted, choose **Yes** to restart.
10. After the reboot, launch **VMware Workstation Pro** from the Start menu.
11. On first launch you may be asked how you intend to use the product. Choose the free personal or commercial use option. **No license key is required.**

[SCREENSHOT REQUIRED]
Capture: The VMware Workstation Pro Custom Setup screen during installation on Windows
Show: The installation path field and both optional checkboxes, with the console tools PATH option ticked
Caption: Figure 5-1: Enabling the console tools during Workstation installation

5.4 Verify

Open an ordinary Command Prompt or PowerShell window, not an elevated one, and run:

```
vmrun -T ws list
```

VERIFY

The command returns Total running VMs: 0 rather than "not recognized". That confirms both the install and the PATH entry.

In Windows, open Network Connections. You should see **VMware Network Adapter VMnet1** and **VMware Network Adapter VMnet8**.

The Workstation window opens with a Home tab offering "Create a New Virtual Machine".

5.5 Troubleshooting

Symptom	Likely cause	Fix
"VMware Workstation and Device/Credential Guard are not compatible"	Hyper-V or VBS is active	Apply the disable steps in 5.2 and reboot; check Group Policy if it re-enables itself
"This host supports Intel VT-x, but VT-x is disabled"	Firmware setting off	Reboot into UEFI/BIOS and enable Intel VT-x or AMD SVM Mode
Installer rolls back near the end	Antivirus blocking driver installation, or a pending reboot from a prior update	Reboot, temporarily pause the endpoint agent, and reinstall as administrator
VMnet adapters missing	Virtual network editor did not complete	Run the installer again and choose Repair, or open Edit > Virtual Network Editor and click Restore Defaults
Guests run but very slowly	Running nested on Hyper-V	Confirm hypervisorlaunchtype is off and that Memory Integrity is disabled

6. Installing VMware Workstation on Fedora

6.1 Explain

The Linux installer is a self-extracting .bundle file. It unpacks the product, then compiles two kernel modules against your running kernel: vmmon, the virtual machine monitor, and vmnet, the virtual networking layer.

That compilation step is the entire story of installing VMware on Fedora. Fedora ships new kernels frequently, sometimes faster than VMware publishes module compatibility patches, and every kernel update requires the modules to be rebuilt. Plan for it rather than being surprised by it.

> **NOTE**
> These instructions target **Fedora Workstation 44** with a GNOME desktop. Workstation Pro needs a graphical session for its console. If you are running Fedora Server, either add a desktop environment or use a different host.

6.2 Prerequisites

Install the build dependencies before you run the installer, not after it fails:

```
sudo dnf update -y
sudo reboot

# After the reboot, install build tools and headers for the RUNNING kernel:
sudo dnf install -y kernel-devel-$(uname -r) kernel-headers-$(uname -r) \
  gcc gcc-c++ make elfutils-libelf-devel perl python3 libxcrypt-compat

# Confirm the headers match the running kernel:
uname -r
ls /usr/src/kernels/
```

> **WARNING**
> The two commands above must print the same version string. If uname -r shows a newer kernel than the directory under /usr/src/kernels/, you updated the kernel and did not reboot. Reboot first. Building modules against headers for a kernel you are not running produces modules that will not load.

Secure Boot

If Secure Boot is enabled, the kernel will refuse to load unsigned modules, and the VMware modules are unsigned by default. You have three options:

- Disable Secure Boot in UEFI firmware. Simplest, and acceptable on a dedicated lab machine.
- Generate a Machine Owner Key, enroll it with mokutil, and sign vmmon and vmnet after every rebuild. Correct, but it becomes a recurring chore.
- Use a different hypervisor on this host. Fedora includes KVM/QEMU with GNOME Boxes and virt-manager, which are signed and integrated.

```
# Check Secure Boot state:
mokutil --sb-state
```

6.3 Perform

1. Open a terminal and change to the directory holding the downloaded bundle.
2. Make the bundle executable:

   ```
   chmod +x VMware-Workstation-Full-26H1-*.x86_64.bundle
   ```

3. Run the installer with root privileges:

   ```
   sudo ./VMware-Workstation-Full-26H1-*.x86_64.bundle
   ```

   A graphical installer window opens. If you prefer a console install, append --console.

4. Accept both license agreements. There are two: one for Workstation and one for VMware OVF Tool.
5. Choose whether to check for product updates on startup. Say no in a lab.
6. Choose whether to join the Customer Experience Improvement Program. Say no in a lab.
7. Enter the user who will initially connect to the Workstation Server, which is normally your own username.
8. Accept the default shared virtual machines directory, usually /var/lib/vmware/Shared VMs, and the default HTTPS port of 443.
9. Leave the license key field empty and continue. Workstation Pro is free and requires no key.
10. Click **Install** and wait for the bundle to extract.
11. Launch Workstation the first time from a terminal so you can see any module errors:

    ```
    vmware
    ```

    The first launch triggers the kernel module build. You will be prompted for your password so the installer can compile and load vmmon and vmnet.

[SCREENSHOT REQUIRED]
Capture: The VMware Kernel Module Updater dialog on Fedora during first launch
Show: The progress bar building vmmon and vmnet, with the authentication prompt visible
Caption: Figure 6-1: Compiling the vmmon and vmnet kernel modules on first launch

6.4 Verify

```
# Both modules should be listed:
lsmod | grep -E "vmmon|vmnet"

# Services should be active:
systemctl status vmware.service --no-pager

# Version check:
vmware --version
```

VERIFY

lsmod lists both vmmon and vmnet.

vmware --version reports the 26H1 build.

The Workstation window opens with the Home tab available.

6.5 Rebuilding Modules After a Kernel Update

This is the maintenance task that defines running VMware on Fedora. After any kernel update, the modules must be rebuilt against the new kernel or Workstation will refuse to start a virtual machine.

```
# After a kernel update and reboot:
sudo dnf install -y kernel-devel-$(uname -r)
sudo vmware-modconfig --console --install-all

# Confirm:
lsmod | grep -E "vmmon|vmnet"
```

TIP

If vmware-modconfig fails against a very new Fedora kernel, the community project mkubecek/vmware-host-modules on GitHub maintains patches ahead of official support. Check that the branch matches your Workstation version before using it, and treat it as a stopgap rather than a permanent arrangement.

6.6 Troubleshooting

Symptom	Likely cause	Fix
"Unable to install all modules"	Missing or mismatched kernel headers	Install kernel-devel-$(uname -r), confirm it matches uname -r, rerun vmware-modconfig
Modules build but will not load	Secure Boot rejecting unsigned modules	Disable Secure Boot or sign the modules with an enrolled MOK
vmware command not found	Installer did not complete	Rerun the bundle; check /usr/lib/vmware exists
Compile errors after a Fedora upgrade	Kernel newer than VMware support	Boot the previous kernel from the GRUB menu, or apply community host-module patches
Workstation starts but has no network	vmnet not loaded, or services stopped	sudo systemctl restart vmware.service then verify with lsmod

7. Creating a Windows Virtual Machine with VMware Workstation

7.1 Explain

Workstation offers two paths through the new VM wizard. **Typical** asks a handful of questions and picks sane defaults. **Custom** exposes firmware type, controller types, and disk options. Use Custom. Not because Typical is wrong, but because a security student should see the decisions being made on their behalf.

One option deserves a decision before you begin: Easy Install. If you supply a Windows product key and account details, Workstation writes an unattended answer file and installs Windows without interaction. It is fast, and it is exactly wrong for teaching, because students never see the Windows installer. Turn it off.

7.2 Prerequisites

- Windows Server 2025 evaluation ISO in your ISO library (Section 3.4).
- At least 4 GB of RAM free on the host beyond the host reserve.
- At least 40 GB free on the target volume, even though you will declare 80 GB thin.

7.3 Perform

1. In Workstation, choose **File > New Virtual Machine**, or click "Create a New Virtual Machine" on the Home tab.
2. Select **Custom (advanced)** and click **Next**.
3. Leave hardware compatibility at the newest version offered and click **Next**.

   Lower this only if you need to open the VM in an older VMware product, such as an ESXi host running an earlier release.
4. On the guest operating system installation screen, choose **I will install the operating system later**, then **Next**.

   This is the step that avoids Easy Install. It creates the VM with a blank disk and no attached media, and you will attach the ISO manually in step 12. Do it this way.
5. For guest operating system select **Microsoft Windows**, and for version select **Windows Server 2025** (or the closest listed version). Click **Next**.

   This selection is not cosmetic. It sets the default firmware type, the SCSI controller, the virtual NIC model, and which VMware Tools package gets offered.
6. Name the virtual machine something that will still make sense in six months, for example LAB-DC01-WS2025. Set the location to your VM library folder, then click **Next**.

> **WARNING**
> The VM folder must not be inside OneDrive, Dropbox, iCloud Drive, or any other sync folder. Sync clients will attempt to upload multi-gigabyte disk files while the VM is running, which causes lock contention, corruption, and a very confusing failure.

7. On the firmware screen choose **UEFI**, and tick **Secure Boot**.

   Windows Server 2025 expects UEFI. Secure Boot in the guest is realistic and it is what production looks like. You can disable it later per VM if a lab exercise needs unsigned drivers.

8. Set processors. Choose **1 processor with 2 cores**, which presents 2 vCPUs to the guest. Click **Next**.

9. Set memory to **4096 MB**. Click **Next**.

   The wizard shows recommended and maximum values. Ignore the maximum; it reflects host RAM, not host RAM minus what the host needs.

10. For network type choose **Use network address translation (NAT)** for now. Click **Next**.

    NAT gives the guest outbound internet access for updates while keeping it off your physical LAN. Section 23 covers when to change this.

11. Accept the defaults for I/O controller type (**LSI Logic SAS**) and disk type (**NVMe** on modern hardware, or SCSI). Click **Next** through both.

12. Choose **Create a new virtual disk**, set the size to **80 GB**, and select **Store virtual disk as a single file**. Leave "Allocate all disk space now" unticked. Click **Next**, then **Next** again to accept the disk filename.

    A single file performs better than split files on any modern filesystem. Split files exist for FAT32 compatibility, which you no longer have.

13. On the summary screen, click **Customize Hardware**.

14. Select **New CD/DVD (SATA)**. Choose **Use ISO image file**, browse to your Windows Server 2025 ISO, and confirm **Connect at power on** is ticked.

15. While you are in the hardware editor, add a **Trusted Platform Module** if your curriculum needs BitLocker or Credential Guard exercises. Workstation will require the VM to be encrypted before a TPM can be added; follow the prompt and set a password you record.

16. Close the hardware editor and click **Finish**.

17. Click **Power on this virtual machine**. Click inside the console window and press a key quickly when prompted, otherwise the VM will fall through to the UEFI shell.

    If you land at an EFI shell prompt, type exit, choose Boot Manager, and select the CD/DVD device.

18. Work through the Windows Server installer: language, keyboard, **Install now**.

19. Choose an edition. **Windows Server 2025 Standard Evaluation (Desktop Experience)** is the right choice for a teaching lab; Server Core has no GUI.

20. Accept the license terms, choose **Custom: Install Windows only**, select the unallocated 80 GB disk, and click **Next**.

21. Wait for the installation and the reboots. Set the Administrator password when prompted. Record it.

22. When the desktop appears, install VMware Tools: menu **VM > Install VMware Tools**. This mounts a virtual disc inside the guest.

23. Inside the guest, open File Explorer, open the mounted VMware Tools disc, run setup64.exe, choose a Typical installation, and reboot the guest.
24. After the reboot, confirm the guest is fully updated (Settings > Windows Update), then shut the guest down cleanly.
25. With the VM powered off, take a snapshot: **VM > Snapshot > Take Snapshot**, named Clean Build with a description of the date and patch level.

[SCREENSHOT REQUIRED]

Capture: The VMware Workstation Customize Hardware dialog before first power on

Show: The CD/DVD device set to Use ISO image file with the Windows Server 2025 ISO selected and Connect at power on ticked

Caption: Figure 7-1: Attaching the Windows Server evaluation ISO to the new virtual machine

7.4 Why VMware Tools Matters

Students routinely skip this and then complain the VM is unusable. VMware Tools installs paravirtualized drivers and integration services. Without it you get a generic display driver at a fixed low resolution, no clipboard sharing, no drag and drop, poor mouse tracking, no time synchronization, and no graceful shutdown from the host. With it, the VM behaves like a normal machine.

7.5 Verify

VERIFY

The guest reaches the Windows Server desktop and reports the correct edition under System.

The console window resizes and the guest resolution follows it, which proves Tools is loaded.

Clipboard text copies between host and guest.

The guest can reach the internet: open a browser or run Test-NetConnection www.microsoft.com -Port 443 in the guest.

A snapshot named Clean Build appears in **VM > Snapshot > Snapshot Manager**.

7.6 Troubleshooting

Symptom	Likely cause	Fix
Boots to an EFI shell	Missed the "press any key" window	Type exit, use Boot Manager, select the CD/DVD device; or power off and on and press a key faster
Installer cannot see any disk	Controller type the installer has no driver for	Power off, change the disk controller to LSI Logic SAS or NVMe in hardware settings, retry
"This PC can't run Windows 11" (client builds)	No TPM or Secure Boot	Add a TPM device (requires VM encryption) and enable Secure Boot on the firmware page
Screen stuck at low resolution	VMware Tools not installed	VM > Install VMware Tools, run setup64.exe in the guest, reboot

Symptom	Likely cause	Fix
No network inside the guest	Adapter disconnected, or host VMnet services stopped	Check "Connected" on the network adapter; on the host restart the VMware NAT and DHCP services

8. Creating a Linux Virtual Machine with VMware Workstation

8.1 Explain

The mechanics mirror Section 7. Three things differ, and they are the things worth teaching.

- Linux guests need far less memory and disk than Windows guests for the same lab value.
- open-vm-tools from the distribution repository has replaced the VMware Tools package for Linux guests. Install it from the distribution, not from the VMware menu.
- Workstation offers Easy Install for Linux too, and it should be declined for the same reason: students should see the installer.

This walkthrough uses **Ubuntu Server 26.04 LTS**, with a Fedora Server variation at the end.

8.2 Prerequisites

- Ubuntu Server 26.04 LTS ISO, amd64 build, verified by checksum.
- 2 GB of free host RAM beyond the host reserve.
- 20 GB of free space on the target volume.

8.3 Perform

1. Choose **File > New Virtual Machine**, select **Custom (advanced)**, and click **Next**.
2. Accept the newest hardware compatibility and click **Next**.
3. Select **I will install the operating system later** and click **Next**. This declines Easy Install.
4. For guest operating system choose **Linux**, and for version choose **Ubuntu 64-bit**. Click **Next**.

   For Fedora choose **Fedora 64-bit**. If your exact release is not listed, pick the nearest one; the setting drives defaults, not compatibility.
5. Name it LAB-UBSRV01 and set the location inside your VM library. Click **Next**.
6. Choose **UEFI** firmware. Leave Secure Boot ticked; Ubuntu and Fedora both ship signed shims and boot cleanly under Secure Boot.
7. Set **1 processor with 2 cores**. Click **Next**.
8. Set memory to **2048 MB**. Click **Next**.

   Use 4096 MB if you are installing a graphical desktop such as Fedora Workstation or Ubuntu Desktop.
9. Choose **NAT** networking. Click **Next**.
10. Accept the default I/O controller and choose **NVMe** for the disk type where offered. Click **Next**.
11. Create a new virtual disk of **40 GB**, stored as a single file, without preallocating. Click **Next**, then **Next**.
12. On the summary screen click **Customize Hardware**, select the CD/DVD device, choose **Use ISO image file**, and browse to the Ubuntu Server ISO. Confirm **Connect at power on**.

13. Close the hardware editor and click **Finish**, then power on the virtual machine.
14. At the GRUB menu, select **Try or Install Ubuntu Server** and press Enter.
15. Work through the Subiquity installer:

    Language and keyboard layout.

    Installation type: choose the default **Ubuntu Server** rather than the minimized variant.

    Network: accept DHCP. The NAT adapter will pick up an address automatically.

    Proxy and mirror: leave blank and accept the default.

    Storage: choose **Use an entire disk**. Enabling LVM is optional but useful for later exercises in volume management.

    Profile: set your name, the server name, a username, and a password. Record them.

    SSH: tick **Install OpenSSH server**. You will want this immediately.

    Snaps: skip them all for a lab build.
16. Wait for installation to finish, then choose **Reboot Now**. When prompted to remove the installation medium, press Enter.

    If the VM boots back into the installer, power it off, edit the CD/DVD device, and untick **Connect at power on**.
17. Log in at the console and install the guest tools and updates:

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y open-vm-tools

# Only if you installed a graphical desktop:
# sudo apt install -y open-vm-tools-desktop

sudo systemctl enable --now open-vm-tools
ip addr show
sudo reboot
```

18. After the reboot, confirm networking and note the guest IP address, then shut down cleanly with sudo shutdown -h now.
19. Take a snapshot named Clean Build.

8.4 Fedora Server Variation

Everything above holds, with these changes:

Step	Ubuntu Server 26.04	Fedora Server 44
Guest OS version in wizard	Ubuntu 64-bit	Fedora 64-bit
Installer	Subiquity, text mode	Anaconda, graphical
Disk layout	Guided, LVM optional	Automatic, uses Btrfs or LVM by default

Step	Ubuntu Server 26.04	Fedora Server 44
Root account	Disabled, use sudo	Anaconda offers a root password and a user account; create the user and grant it administrator
Guest tools	sudo apt install open-vm-tools	sudo dnf install open-vm-tools
Update command	sudo apt update && sudo apt upgrade -y	sudo dnf upgrade --refresh -y
Firewall	ufw, inactive by default	**`firewalld`, active by default.** Expect to open ports explicitly.
Mandatory access control	AppArmor	**SELinux enforcing.** Denials appear in /var/log/audit/audit.log.

```
# Fedora Server post-install:
sudo dnf upgrade --refresh -y
sudo dnf install -y open-vm-tools
sudo systemctl enable --now vmtoolsd

# Useful early checks for students:
sudo firewall-cmd --list-all
getenforce
ip addr show
```

8.5 Verify

VERIFY

The guest boots to a login prompt without the installation ISO attached.

ip addr show reports an address on the VMware NAT subnet, typically 192.168.x.x.

systemctl status open-vm-tools (Ubuntu) or systemctl status vmtoolsd (Fedora) shows active and running.

You can SSH into the guest from the host using the address the guest reported.

A Clean Build snapshot exists.

8.6 Troubleshooting

Symptom	Likely cause	Fix
Boots back into the installer every time	ISO still attached and connected at power on	Power off, edit CD/DVD device, untick Connect at power on
No IP address in the guest	Adapter not connected, or host DHCP service stopped	Tick Connected on the adapter; restart VMware DHCP and NAT services on the host
Console text is tiny or the window will not resize	open-vm-tools missing	Install open-vm-tools (and open-vm-tools-desktop for GUI guests) and reboot
Cannot SSH from the host to	firewalld blocking, or sshd	sudo systemctl enable --now sshd and sudo

Symptom	Likely cause	Fix
a Fedora guest	not enabled	firewall-cmd --add-service=ssh --permanent && sudo firewall-cmd --reload
Guest hangs at boot after a kernel update	Snapshot chain or disk full on host	Check host free space; revert to Clean Build and reapply updates in stages

Part 3: Oracle VirtualBox

VirtualBox is the most portable option in this guide. The same product, with the same interface and the same VM format, runs on Windows, Linux, macOS, and Solaris. For a class where students arrive with a mixture of hardware, that consistency is worth a great deal.

It is also open source. The base package is licensed under GPLv3, which means no account, no registration form, and no compliance screening between a student and the download. The current line is **VirtualBox 7.2**, with 7.2.16 released on 18 August 2026.

One licensing subtlety worth teaching

The base package is GPL. The **Extension Pack** is not. It is proprietary, distributed under the Personal Use and Educational License, and free only for personal and educational use. Commercial use of the Extension Pack requires a paid licence from Oracle, and Oracle has audited organizations over exactly this.

The Extension Pack adds USB 2.0 and 3.0 device support, VirtualBox RDP, disk image encryption, PXE boot for Intel cards, and NVMe support. In a classroom you are covered by the educational terms. Teach students to check before they install it at work.

9. Downloading VirtualBox from the Oracle Website

9.1 Explain

There are two official download pages and both are legitimate. The virtualbox.org page is the community-facing site and is usually the fastest route. The oracle.com page carries the same binaries.

```
Community download page (recommended):
https://www.virtualbox.org/wiki/Downloads

Oracle download page:
https://www.oracle.com/virtualization/technologies/vm/downloads/virtualbox-
downloads.html

Older builds:
https://www.virtualbox.org/wiki/Download_Old_Builds
```

> **WARNING**
> Download only from those two domains. VirtualBox is heavily impersonated by download aggregators that wrap the installer in adware bundlers. If a page offers a "download manager" or a "safe downloader", you are on the wrong site.

9.2 Perform

1. Open https://www.virtualbox.org/wiki/Downloads.
2. Under **VirtualBox platform packages**, pick the host you are installing on, not the guest you plan to run:

Windows hosts for a Windows PC.

Linux distributions for Fedora, then choose the matching Fedora build.

macOS / Intel hosts for an Intel Mac. **macOS / Apple Silicon hosts** for an M-series Mac.

3. Download the platform package.
4. Download the **VirtualBox Extension Pack** from the same page. There is a single Extension Pack file that works on all host platforms.
5. Download the **SHA256 checksums** file linked on the same page.
6. Verify the platform package before installing it:

```
# Windows PowerShell
Get-FileHash -Algorithm SHA256 .\VirtualBox-7.2.16-*-Win.exe

# Linux
sha256sum VirtualBox-7.2.16*.rpm

# macOS
shasum -a 256 VirtualBox-7.2.16-*-OSX.dmg
```

7. Compare the output against the published SHA256SUMS file. They must match exactly.

[SCREENSHOT REQUIRED]

Capture: The virtualbox.org Downloads page

Show: The platform packages list with Windows hosts, Linux distributions, and both macOS options visible, plus the Extension Pack link below

Caption: Figure 9-1: Choosing the correct VirtualBox platform package for your host

9.3 Verify

VERIFY

You have a platform installer named for your host, for example VirtualBox-7.2.16-<build>-Win.exe.

You have Oracle_VirtualBox_Extension_Pack-7.2.16.vbox-extpack.

The Extension Pack version matches the platform package version exactly. A mismatched pack will refuse to install.

Your computed SHA256 matches the published value.

10. Installing VirtualBox on Windows

10.1 Explain

The Windows installer registers kernel drivers (VBoxDrv, plus the networking filter drivers) and installs the VirtualBox Host-Only Network adapter. It also warns you that it will briefly reset your network connections, which is real: the adapter installation drops active connections for a few seconds.

VirtualBox coexists with Hyper-V better than VMware does, but "coexists" is not "performs". When Hyper-V is active, VirtualBox runs through the Windows Hypervisor Platform API, and guests are measurably slower. The same conflict check from Section 5.2 applies here.

10.2 Prerequisites

- Windows 10 or Windows 11, 64-bit.
- VT-x or AMD-V enabled in firmware.
- Administrator rights on the host.
- The Microsoft Visual C++ Redistributable, which recent installers bundle automatically.

> **NOTE**
> If you also run VMware Workstation on the same host, you can install both. Do not run virtual machines in both products at the same time. Two Type 2 hypervisors competing for the same CPU extensions produce host instability, and on some builds a bug check.

10.3 Perform

1. Right-click the installer and choose **Run as administrator**.
2. On the welcome screen click **Next**.
3. On Custom Setup, leave all features enabled and accept the install path, then click **Next**.
4. The next screen asks about shortcuts and file associations. Leave **Register file associations** ticked so .vbox and .ova files open in VirtualBox. Click **Next**.
5. A warning appears: **Network Interfaces will be reset**. Close anything network-dependent, then click **Yes**.
6. If the installer offers to install missing dependencies such as the Python bindings or the Visual C++ runtime, accept.
7. Click **Install**. Approve any Windows driver installation prompts from Oracle Corporation.
8. Click **Finish** and let VirtualBox launch.

Install the Extension Pack

9. In VirtualBox Manager choose **File > Tools > Extension Pack Manager**, then click **Install**.
   Alternatively, simply double-click the downloaded .vbox-extpack file.
10. Browse to the downloaded Extension Pack file and select it.

11. Read the Personal Use and Educational License and click **I Agree**. Approve the elevation prompt.

WARNING

This is the licence discussed at the start of Part 3. Educational and personal use is free. Commercial use requires a paid Oracle licence. Read the screen instead of clicking past it, and make your students read it too.

[SCREENSHOT REQUIRED]

Capture: The VirtualBox Extension Pack Manager after a successful installation

Show: The Oracle VirtualBox Extension Pack listed with a version number matching the base package

Caption: Figure 10-1: Confirming the Extension Pack version matches the base installation

10.4 Verify

```
VBoxManage --version
VBoxManage list extpacks
VBoxManage list hostinfo
```

If VBoxManage is not recognized, add C:\Program Files\Oracle\VirtualBox to your PATH, or run the command from that directory.

VERIFY

VBoxManage --version prints 7.2.16 or later.

VBoxManage list extpacks lists the Oracle VirtualBox Extension Pack, **Usable: true**.

Network Connections shows a **VirtualBox Host-Only Network** adapter.

VirtualBox Manager opens with an empty machine list.

10.5 Troubleshooting

Symptom	Likely cause	Fix
"Kernel driver not installed (rc=-1908)"	Driver failed to load or was blocked	Reinstall as administrator; check Secure Boot and endpoint protection driver blocking
Extension Pack refuses to install	Version mismatch with the base package	Download the Extension Pack matching your exact base version
Only 32-bit guest options available	VT-x/AMD-V disabled, or Hyper-V holding the extensions	Enable virtualization in firmware; apply the Hyper-V steps from Section 5.2
Guests very slow, host fine	Running on the Windows Hypervisor Platform	Disable Hyper-V, Memory Integrity, and WSL2, then reboot
Host network drops during install	Expected, the filter drivers reset adapters	Wait 30 seconds; if it persists, disable and re-enable the physical adapter

11. Installing VirtualBox on Fedora

11.1 Explain

You have two routes on Fedora. The Oracle RPM is a single download and matches the version on the website. The Oracle DNF repository is slightly more work to set up once and then keeps VirtualBox updated with the rest of the system, which is the better choice for a machine you will maintain.

Either way, VirtualBox builds a kernel module (vboxdrv) against your running kernel. Unlike VMware, VirtualBox uses **DKMS**, which rebuilds the module automatically whenever the kernel changes. That single difference makes VirtualBox meaningfully less trouble on Fedora than VMware Workstation.

11.2 Prerequisites

```
sudo dnf upgrade --refresh -y
sudo reboot

# After reboot, install build prerequisites:
sudo dnf install -y @development-tools dkms \
  kernel-devel-$(uname -r) kernel-headers-$(uname -r) \
  elfutils-libelf-devel qt5-qtx11extras libxkbcommon-devel

# Confirm the headers match the running kernel:
uname -r && ls /usr/src/kernels/
```

Secure Boot on Fedora

Fedora enables Secure Boot on most systems, and an unsigned vboxdrv will not load. Check the state:

```
mokutil --sb-state
```

If Secure Boot is enabled, you can either disable it in firmware, or sign the module with your own Machine Owner Key. The signing route is more work but it is the right habit, and it is a good teaching exercise in its own right:

```
# Generate a Machine Owner Key
sudo mkdir -p /root/mok && cd /root/mok
sudo openssl req -new -x509 -newkey rsa:2048 -keyout MOK.priv \
  -outform DER -out MOK.der -nodes -days 36500 \
  -subj "/CN=VirtualBox Lab Module Signing/"

# Enroll it. You will set a one-time password and confirm at the next boot.
sudo mokutil --import MOK.der
sudo reboot

# After enrolling, sign the module (repeat after each rebuild):
sudo /usr/src/kernels/$(uname -r)/scripts/sign-file sha256 \
  /root/mok/MOK.priv /root/mok/MOK.der \
  $(modinfo -n vboxdrv)
sudo modprobe vboxdrv
```

11.3 Perform: Option A, the Oracle DNF Repository (recommended)

1. Add the Oracle repository:

```
sudo dnf config-manager addrepo \
  --from-repofile=https://download.virtualbox.org/virtualbox/rpm/fedora/
virtualbox.repo
```

On older Fedora releases the syntax is sudo dnf config-manager --add-repo https://download.virtualbox.org/virtualbox/rpm/fedora/virtualbox.repo.

2. Import the Oracle signing key:

```
sudo rpm --import https://www.virtualbox.org/download/oracle_vbox_2016.asc
```

3. Install the package:

```
sudo dnf install -y VirtualBox-7.2
```

4. Build and load the kernel module:

```
sudo /sbin/vboxconfig
```

5. Add your user to the vboxusers group so USB passthrough works:

```
sudo usermod -aG vboxusers $USER

# Log out and back in, then confirm:
groups
```

11.4 Perform: Option B, the Standalone RPM

1. Download the Fedora RPM from the VirtualBox downloads page.
2. Install it, letting DNF resolve dependencies:

```
sudo dnf install -y ./VirtualBox-7.2-7.2.16_*_fedora*.x86_64.rpm
```

3. Build the kernel module and add yourself to vboxusers exactly as in Option A steps 4 and 5.

11.5 Install the Extension Pack

1. Download the Extension Pack matching your installed version.
2. Install it from the command line:

```
sudo VBoxManage extpack install \
  Oracle_VirtualBox_Extension_Pack-7.2.16.vbox-extpack
```

Read and accept the licence when prompted. To replace an existing pack, add --replace.

11.6 Verify

```
VBoxManage --version
VBoxManage list extpacks

# Module loaded?
lsmod | grep vboxdrv

# Service healthy?
systemctl status vboxdrv --no-pager
```

VERIFY

lsmod shows vboxdrv.

systemctl status vboxdrv reports active.

groups includes vboxusers.

VirtualBox launches from the Activities overview.

11.7 After a Kernel Update

DKMS normally handles this without intervention. If a guest refuses to start after a kernel update, rebuild manually:

```
sudo dnf install -y kernel-devel-$(uname -r)
sudo /sbin/vboxconfig
lsmod | grep vboxdrv

# With Secure Boot enabled, re-sign after every rebuild:
sudo /usr/src/kernels/$(uname -r)/scripts/sign-file sha256 \
  /root/mok/MOK.priv /root/mok/MOK.der $(modinfo -n vboxdrv)
sudo modprobe vboxdrv
```

11.8 Troubleshooting

Symptom	Likely cause	Fix
"Kernel driver not installed (rc=-1908)"	Module not built or not loaded	Run sudo /sbin/vboxconfig and read the output carefully
"Required key not available" on modprobe	Secure Boot rejecting an unsigned module	Enroll a MOK and sign vboxdrv, or disable Secure Boot
USB devices never appear in the guest	User not in vboxusers, or Extension Pack missing	sudo usermod -aG vboxusers $USER, log out and back in, confirm the pack is installed
Build fails with missing headers	Kernel updated without a reboot	Reboot, install kernel-devel-$(uname -r), rerun vboxconfig
GUI will not start, Qt errors	Missing Qt dependencies	Install qt5-qtx11extras and libxkbcommon-devel, then relaunch

12. Creating a Windows Virtual Machine with VirtualBox

12.1 Explain

VirtualBox 7 introduced **unattended installation**. When you attach a Windows ISO in the new VM wizard, VirtualBox detects it and offers to install Windows automatically using an answer file it generates. Like VMware Easy Install, it is convenient and it is the wrong choice for teaching. There is a checkbox to skip it, and this walkthrough uses it.

The other VirtualBox-specific decision is the default virtual disk size, which is far too small for a modern Windows Server. Change it in the wizard rather than resizing later.

12.2 Prerequisites

- Windows Server 2025 evaluation ISO.
- 4 GB of host RAM free beyond the host reserve.
- 40 GB free on the target volume.
- Extension Pack installed, if you want USB 2.0/3.0 passthrough.

12.3 Perform

1. In VirtualBox Manager click **New**.
2. Set the name to LAB-DC01-WS2025. Set the machine folder to your VM library. VirtualBox will guess the type and version from the name; correct them if needed to **Microsoft Windows** and **Windows 2025 (64-bit)** or the nearest listed option.
3. In the **ISO Image** field, select your Windows Server ISO.
4. **Tick "Skip Unattended Installation".**

   This is the important click in this procedure. Without it, VirtualBox generates an answer file and installs Windows without showing students a single installer screen.
5. Expand **Hardware**:

   Base Memory: **4096 MB**.

   Processors: **2**.

   Tick **Enable EFI (special OSes only)**. Windows Server 2025 expects UEFI.
6. Expand **Hard Disk**:

   Choose **Create a Virtual Hard Disk Now**.

   Set the size to **80 GB**. The default of 50 GB is not enough for a patched Windows Server with roles installed.

   Leave **Pre-allocate Full Size** unticked so the disk stays thin.

   Leave the format as VDI unless you need portability to another product.
7. Click **Finish**. The VM appears in the list but is not yet running.
8. With the VM selected, click **Settings** and make these adjustments before first boot:

System > Motherboard: confirm Enable EFI is ticked. If your curriculum needs BitLocker or Credential Guard, set **TPM** to **v2.0** and enable **Secure Boot**.

System > Processor: confirm 2 CPUs, and tick **Enable Nested VT-x/AMD-V** only if you plan to run a hypervisor inside this guest.

Display > Screen: raise **Video Memory** to **128 MB**. The default is low and produces a poor console.

Network > Adapter 1: set **Attached to: NAT** for now.

Storage: confirm the ISO is attached to the optical drive under the IDE or SATA controller.

9. Click **OK**, then **Start**.
10. Press a key quickly when the "Press any key to boot from CD or DVD" prompt appears.

    If you land at an EFI shell, type exit, choose Boot Manager, and select the optical device.
11. Complete the Windows Server installer: language, **Install now**, choose **Standard Evaluation (Desktop Experience)**, accept the licence, choose **Custom**, select the 80 GB disk, and click **Next**.
12. Set the Administrator password when prompted and record it.
13. Install **Guest Additions**. From the VM window menu choose **Devices > Insert Guest Additions CD image**.
14. Inside the guest, open File Explorer, open the mounted disc, run VBoxWindowsAdditions.exe, accept the driver prompts from Oracle Corporation, and reboot the guest.
15. After the reboot, apply Windows Updates, shut the guest down cleanly, and take a snapshot named Clean Build from **Machine > Take Snapshot**.

[SCREENSHOT REQUIRED]
Capture: The VirtualBox Create Virtual Machine wizard with the Windows Server ISO selected
Show: The ISO Image field populated and the "Skip Unattended Installation" checkbox ticked
Caption: Figure 12-1: Skipping unattended installation so students see the Windows installer

12.4 Verify

VERIFY

The guest reaches the Windows Server desktop.

Resizing the VM window changes the guest resolution, which proves Guest Additions is loaded.

Bidirectional clipboard works after enabling it under **Devices > Shared Clipboard > Bidirectional**.

The guest reaches the internet through the NAT adapter.

Clean Build appears in the Snapshots view.

12.5 Troubleshooting

Symptom	Likely cause	Fix
Only 32-bit versions listed at VM creation	VT-x/AMD-V unavailable to VirtualBox	Enable virtualization in firmware; disable Hyper-V, Memory Integrity, and WSL2
"Windows cannot be installed to this disk"	MBR/GPT mismatch with firmware setting	Enable EFI in System > Motherboard, delete the partition, and let the installer recreate it
Windows setup complains about TPM or Secure Boot (client builds)	Both are off by default in VirtualBox	Set TPM to v2.0 and enable Secure Boot under System > Motherboard
Guest Additions installs but resolution stays fixed	Video memory too low, or reboot skipped	Raise video memory to 128 MB and reboot the guest
Mouse pointer is trapped in the VM window	Guest Additions not installed yet	Press the Host key (right Ctrl by default) to release, then install Guest Additions
VM aborts immediately on start	Kernel driver or Hyper-V conflict	Check VBoxManage list hostinfo; verify vboxdrv is loaded on Linux, or disable Hyper-V on Windows

13. Creating a Linux Virtual Machine with VirtualBox

13.1 Explain

Linux guests in VirtualBox are the least troublesome combination in this entire guide. The main thing worth knowing is that Guest Additions on Linux is compiled inside the guest, so the guest needs build tools before you run the installer. Skipping that produces a "modules failed to build" message that students misread as a broken installation.

This walkthrough uses **Ubuntu Server 26.04 LTS**, with Fedora Server notes at the end.

13.2 Perform

1. In VirtualBox Manager click **New**.
2. Name it LAB-UBSRV01, set the folder, and confirm the type is **Linux** and version **Ubuntu (64-bit)**.
3. Select the Ubuntu Server ISO in the **ISO Image** field, then **tick "Skip Unattended Installation"**.
4. Set Base Memory to **2048 MB** and Processors to **2**. Tick **Enable EFI**.
5. Create a new virtual hard disk of **40 GB**, thin, VDI format.
6. Click **Finish**, then open **Settings** and adjust:

   Display > Screen: video memory **128 MB**.

   Network > Adapter 1: **NAT**.

   General > Advanced: set Shared Clipboard and Drag'n'Drop to **Bidirectional** if your lab policy permits it.
7. Start the VM and choose **Try or Install Ubuntu Server** at the GRUB menu.
8. Work through the Subiquity installer: language, keyboard, default Ubuntu Server install, DHCP networking, no proxy, default mirror, **Use an entire disk**, profile details, and **Install OpenSSH server**. Skip the snap selection.
9. Reboot when prompted and remove the installation medium if asked.

   If the VM boots into the installer again, power it off and remove the ISO from **Settings > Storage**.
10. Log in and prepare the guest for Guest Additions:

    ```
    sudo apt update && sudo apt upgrade -y
    sudo apt install -y build-essential dkms linux-headers-$(uname -r)
    sudo reboot
    ```
11. After the reboot, from the VM window menu choose **Devices > Insert Guest Additions CD image**, then in the guest:

    ```
    sudo mkdir -p /mnt/cdrom
    sudo mount /dev/cdrom /mnt/cdrom
    sudo /mnt/cdrom/VBoxLinuxAdditions.run
    sudo reboot
    ```
12. After the reboot, add your user to the vboxsf group if you intend to use shared folders:

```
sudo usermod -aG vboxsf $USER
ip addr show
```

13. Shut down cleanly and take a snapshot named Clean Build.

13.3 Fedora Server Variation

```
# Prepare for Guest Additions on Fedora:
sudo dnf upgrade --refresh -y
sudo dnf install -y gcc make perl kernel-devel-$(uname -r) \
  kernel-headers-$(uname -r) dkms elfutils-libelf-devel
sudo reboot

# After reboot, with the Guest Additions image inserted:
sudo mkdir -p /mnt/cdrom
sudo mount /dev/cdrom /mnt/cdrom
sudo /mnt/cdrom/VBoxLinuxAdditions.run
sudo reboot

# Fedora-specific checks worth doing with students:
getenforce                    # SELinux should report Enforcing
sudo firewall-cmd --list-all  # firewalld is active by default
sudo systemctl enable --now sshd
sudo firewall-cmd --add-service=ssh --permanent && sudo firewall-cmd --reload
```

NOTE

SELinux occasionally blocks VirtualBox shared folder mounts on Fedora guests. If a shared folder mounts but is unreadable, check sudo ausearch -m avc -ts recent before assuming the mount failed. Reading the audit log is the lesson here, not the workaround.

13.4 Verify

VERIFY

The guest boots to a login prompt with no ISO attached.

ip addr show reports an address on the VirtualBox NAT range, typically 10.0.2.x.

lsmod | grep vboxguest shows the Guest Additions module loaded.

The console resizes with the window, and the clipboard works in both directions.

A Clean Build snapshot exists.

13.5 Troubleshooting

Symptom	Likely cause	Fix
Guest Additions: "modules failed to build"	Missing kernel headers or compiler	Install build-essential dkms linux-headers-$(uname -r) (Ubuntu) or the kernel-devel set (Fedora), reboot, rerun
Shared folder mounts but is unreadable	User not in vboxsf, or SELinux denial	Add the user to vboxsf and log out; on Fedora check ausearch -m avc -ts recent

Symptom	Likely cause	Fix
Console stuck at 800x600	Guest Additions not loaded, or low video memory	Verify lsmod | grep vboxvideo; raise video memory to 128 MB
No network in the guest	Cable unplugged in adapter settings	Settings > Network > Adapter 1, tick **Cable Connected**
Very slow disk in the guest	Host antivirus scanning the VDI file	Exclude the VM library folder from real-time scanning on the host

Part 4: VMware Fusion (macOS)

VMware Fusion is the macOS member of the same family as Workstation. It shares the virtual machine format, so a .vmwarevm bundle moves between Fusion and Workstation with only the usual guest driver adjustments. Like Workstation, it is free for commercial, educational, and personal use and requires no licence key. The current release is **Fusion Pro 26H1**, published in May 2026.

The Apple Silicon Reality Check

This section decides everything you do in Part 4, so read it before downloading anything.

Apple Silicon Macs (M1 through M5) use ARM64 processors. Fusion on Apple Silicon **virtualizes ARM64 guests only**. It does not emulate x86-64. There is no setting to change this and no workaround inside Fusion.

Your Mac	What Fusion can run	Windows guest	Linux guest
Intel Mac	x86-64 guests	Windows 11 x64, **Windows Server 2025 x64**	Any x86-64 distribution
Apple Silicon (M1-M5)	**ARM64 guests only**	Windows 11 ARM64. **No ARM64 Windows Server evaluation ISO exists.**	ARM64 builds: Ubuntu Server arm64, Fedora aarch64

> **WARNING**
> If your course requires Windows Server specifically and your students are on Apple Silicon, Fusion cannot deliver it locally. Microsoft has not published an ARM64 Windows Server evaluation image. The practical options are: run Windows Server on a Windows or Linux host, run it on a lab server the class connects to, use a cloud instance, or emulate it in UTM and accept that installation may take hours. Decide this before the syllabus is written, not during week three.

14. Downloading VMware Fusion from the Broadcom Website

14.1 Explain

Fusion travels the same road as Workstation: a free Broadcom Support Portal account, a completed profile, an export compliance screening, and a download under the VMware Cloud Foundation division. If you already built an account for Section 4, you can reuse it.

14.2 Prerequisites

- A free Broadcom Support Portal account with a fully completed profile.
- macOS 13 Ventura or later. Check the release notes for the exact minimum for 26H1.
- Roughly 1.5 GB free for the download and installation.

14.3 Perform

1. Open https://www.broadcom.com and select **Support Portal** in the top right.
2. Sign in, or register a free Basic account at https://profile.broadcom.com/web/registration and complete your profile fully, including street address and country.
3. At https://support.broadcom.com, use the division selector and choose **VMware Cloud Foundation**.
4. Select **My Downloads** in the left navigation, then **Free Software Downloads available HERE**.
5. Search for VMware Fusion and open the product entry.

   Ignore any result named Bitfusion. It is a different product entirely.
6. Expand the current release, **VMware Fusion Pro 26H1**, and locate the macOS .dmg.
7. Click the **Terms and Conditions** hyperlink to open the agreement, return to the download tab, tick the agreement checkbox, and start the download.
8. Complete the Trade Compliance form if it appears.

[SCREENSHOT REQUIRED]
Capture: The Broadcom Support Portal showing the VMware Fusion Pro 26H1 download entry
Show: The division set to VMware Cloud Foundation and the macOS .dmg file listed with its build number
Caption: Figure 14-1: Locating the free VMware Fusion Pro download

14.4 Verify

```
shasum -a 256 ~/Downloads/VMware-Fusion-26H1-*.dmg
ls -lh ~/Downloads/VMware-Fusion-26H1-*.dmg
```

VERIFY
You have a .dmg file of roughly 1 GB or more.

A file of a few kilobytes means an error page was saved instead of the installer.

15. Installing VMware Fusion on a MacBook

15.1 Explain

Fusion installs as a normal macOS application, but it needs two things macOS does not grant automatically: a system extension approval, and a set of privacy permissions. Both are approved through System Settings, and both are where students get stuck.

15.2 Perform

1. Double-click the downloaded .dmg to mount it.
2. Double-click the **VMware Fusion** icon in the mounted volume.

   Do not drag it to Applications first. The installer performs the copy and the privileged setup itself.
3. When macOS asks whether you are sure you want to open an application downloaded from the internet, click **Open**.
4. Enter your macOS administrator password when prompted.
5. When asked for a licence, choose the free personal or commercial use option. **No key is required.**
6. Approve the system extension if prompted. macOS will show a message that a system extension from VMware was blocked.

   Open **System Settings > General > Login Items & Extensions** (or **Privacy & Security** on older releases), find the blocked VMware item, and click **Allow**. You may need to reboot.
7. Grant the privacy permissions Fusion requests. Under **System Settings > Privacy & Security**, review:

   Accessibility, so Fusion can capture keyboard and mouse input for guests.

   Screen & System Audio Recording, needed for some display features.

   Files and Folders or **Full Disk Access**, so Fusion can read ISOs and write VM bundles in your chosen location.

   Local Network, so guests can reach other devices on your network in bridged mode.
8. Launch Fusion from Applications. On first run it creates its virtual networking and offers to check for updates.
9. Set your default VM location before creating anything, under **VMware Fusion > Settings > General**. Point it somewhere with space and, critically, somewhere iCloud Drive does not sync.

WARNING

By default, Fusion stores virtual machines in ~/Virtual Machines. If your Mac has Desktop & Documents Folders syncing enabled in iCloud Drive, keep VMs well away from Desktop and Documents. iCloud will attempt to upload a 60 GB disk file while it is being written to, and the result ranges from a stalled Mac to a corrupt virtual machine.

[SCREENSHOT REQUIRED]

Capture: macOS System Settings showing the blocked VMware system extension

Show: The Privacy & Security or Login Items & Extensions pane with the VMware entry and the Allow button

Caption: Figure 15-1: Approving the VMware system extension on macOS

15.3 Verify

```
# Version check:
/Applications/VMware\ Fusion.app/Contents/Library/vmrun -T fusion list

# Optional: add the Fusion tools to your PATH
echo 'export PATH="/Applications/VMware Fusion.app/Contents/Library:$PATH"' \
  >> ~/.zshrc
source ~/.zshrc
vmrun -T fusion list
```

VERIFY

Fusion opens to the Virtual Machine Library window.

vmrun -T fusion list returns Total running VMs: 0.

No blocked-extension warning remains in System Settings.

15.4 Troubleshooting

Symptom	Likely cause	Fix
"System extension blocked"	macOS has not been told to allow it	System Settings > General > Login Items & Extensions, allow the VMware item, then reboot
Keyboard input does not reach the guest	Accessibility permission not granted	System Settings > Privacy & Security > Accessibility, enable VMware Fusion
Fusion cannot see your ISO files	Files and Folders permission missing	Grant Full Disk Access, or move the ISO library to a folder Fusion is allowed to read
Guest cannot reach other devices on the LAN	Local Network permission denied	System Settings > Privacy & Security > Local Network, enable VMware Fusion
Application will not open at all	Quarantine attribute or an incomplete download	Redownload, verify the checksum, and open from the mounted DMG rather than a copy

16. Creating a Windows Virtual Machine with VMware Fusion

16.1 Explain

Which procedure you follow depends on your hardware. Read 16.2 to pick, then follow either 16.3 or 16.4.

16.2 Choose Your Path

Host	Guest to build	Media source	Follow
Intel Mac	Windows Server 2025 x64, or Windows 11 x64	Microsoft Evaluation Center (Section 3.4)	Procedure 16.3
Apple Silicon Mac	**Windows 11 ARM64** (client, not server)	Microsoft Windows 11 ARM64 download, or Fusion's built-in fetch	Procedure 16.4

16.3 Procedure: Windows Server on an Intel Mac

1. In Fusion choose **File > New**.
2. Select **Install from disc or image** and click **Continue**.
3. Drag your Windows Server 2025 ISO onto the window, or click **Use another disc or disc image** and browse to it. Click **Continue**.
4. On the Easy Install screen, **untick "Use Easy Install"**.

   Same reasoning as Sections 7 and 12. Easy Install writes an answer file and hides the Windows installer from your students.
5. On the firmware screen choose **UEFI**. If you need BitLocker or Credential Guard exercises, also enable **UEFI Secure Boot**, which requires encrypting the VM.
6. On the Finish screen click **Customize Settings**, name the VM LAB-DC01-WS2025, and save it into your VM library folder.
7. In the settings window, adjust before first boot:

   Processors & Memory: 2 processor cores, 4096 MB memory.

   Hard Disk: 80 GB, do not pre-allocate, use a single file.

   Network Adapter: **Share with my Mac** (NAT).

   CD/DVD: confirm the ISO is attached and set to connect at power on.
8. Close settings and click the play button to power on.
9. Press a key when prompted to boot from the ISO, then complete the Windows Server installer: edition **Standard Evaluation (Desktop Experience)**, custom install, target the 80 GB disk.
10. Set the Administrator password and record it.

11. Install VMware Tools: **Virtual Machine > Install VMware Tools**, then run setup64.exe from the mounted disc inside the guest and reboot.
12. Apply Windows Updates, shut down cleanly, and take a snapshot named Clean Build from **Virtual Machine > Snapshots**.

16.4 Procedure: Windows 11 ARM64 on an Apple Silicon Mac

Getting ARM64 Windows media

You have two routes. Both are legitimate; the first is easier.

- **Fusion's built-in download.** Recent Fusion releases can fetch and prepare a Windows 11 ARM image for you from within the new virtual machine wizard. If the option appears, it is the least friction.
- **Microsoft's own ARM64 media.** Microsoft publishes Windows 11 ARM64 images on its Windows 11 download pages, and the Windows Insider Program publishes ARM64 preview builds. Search from https://www.microsoft.com/software-download/ rather than following a third-party link.

> **WARNING**
> Do not obtain Windows media from ISO mirror sites, torrent trackers, or "UUP" repackagers unless you fully understand what you are running. In a security course this is not a licensing footnote, it is the lesson: the provenance of your installation media is part of your threat model.

Build the virtual machine

1. In Fusion choose **File > New**.
2. If Fusion offers **Get Windows from Microsoft** or a similar built-in download option, select it, choose Windows 11 ARM64, and let Fusion download and prepare the media. Then skip to step 6.
3. Otherwise select **Install from disc or image** and click **Continue**.
4. Drag your Windows 11 ARM64 ISO onto the window and click **Continue**.
5. Untick **Use Easy Install** and choose **UEFI** firmware. Enable **Secure Boot** and add a **TPM**, both of which Windows 11 requires. Fusion will prompt to encrypt the VM in order to add the TPM; set a password and record it.
6. Click **Customize Settings**, name the VM LAB-W11-ARM, and save it to your VM library.
7. Set **Processors & Memory** to 2 cores and 4096 MB, and the hard disk to 80 GB thin.
8. Power on and complete Windows Setup.
9. If Windows Setup stops on a network requirement and no adapter is detected, press **Shift+F10** at the language screen to open a command prompt, then run:

   ```
   OOBE\BYPASSNRO
   ```

   The VM restarts and setup then offers "I don't have internet" and "Continue with limited setup". Microsoft changes this behavior between builds, so this may or may not be needed.

10. When the desktop appears, install VMware Tools from **Virtual Machine > Install VMware Tools**, run the installer inside the guest, and reboot. This installs the network and display drivers, which is why networking may be missing until you do.

11. Apply Windows Updates, shut down cleanly, and take a Clean Build snapshot.

[SCREENSHOT REQUIRED]

Capture: The VMware Fusion new virtual machine wizard with Easy Install unticked

Show: The Windows ISO attached, the Easy Install checkbox clear, and the UEFI firmware option selected

Caption: Figure 16-1: Declining Easy Install so the Windows installer is visible to students

16.5 Verify

VERIFY

The guest boots to the Windows desktop and reports the expected edition and architecture under Settings > System > About.

On an Apple Silicon host, System Type reads ARM-based processor. That is expected and correct.

Resizing the Fusion window changes the guest resolution, which proves VMware Tools is loaded.

The guest has network connectivity and the clipboard works between host and guest.

A Clean Build snapshot exists.

16.6 Troubleshooting

Symptom	Likely cause	Fix
"The guest operating system is not supported"	x86-64 media on an Apple Silicon host	Obtain ARM64 media, or move the workload to an Intel or PC host
No network during Windows Setup	Drivers arrive with VMware Tools	Use OOBE\BYPASSNRO at the language screen, finish setup offline, then install Tools
Setup refuses to continue: TPM or Secure Boot required	Neither enabled on the VM	Power off, enable Secure Boot and add a TPM (this requires encrypting the VM), retry
x86 applications fail inside an ARM64 guest	Emulation layer limits inside Windows on ARM	Expected. Use ARM64 native tools where possible and treat this as a teaching point about architecture
Very poor graphics performance	VMware Tools missing	Install VMware Tools and reboot the guest

17. Creating a Linux Virtual Machine with VMware Fusion

17.1 Explain

Linux is where Apple Silicon stops being a problem. Ubuntu, Fedora, Debian, and most other major distributions publish ARM64 builds, so an M-series Mac runs Linux guests at full virtualized speed with no emulation and no compromise.

Download the right architecture and everything else in this section is routine.

Host	Ubuntu Server 26.04 LTS	Fedora Server 44
Intel Mac	ubuntu-26.04-live-server-amd64.iso	Fedora-Server-dvd-x86_64-44-*.iso
Apple Silicon Mac	**`ubuntu-26.04-live-server-arm64.iso`**	* `Fedora-Server-dvd-aarch64-44-.iso`**

> **NOTE**
> Ubuntu ARM64 server images are published under https://cdimage.ubuntu.com/releases/. Fedora aarch64 images are on the standard Fedora download pages; look for the aarch64 architecture selector rather than assuming the default link is what you want.

17.2 Perform

1. In Fusion choose **File > New**, select **Install from disc or image**, and click **Continue**.
2. Drag the Ubuntu Server ISO onto the window, matching your host architecture, and click **Continue**.
3. If a Linux Easy Install screen appears, **untick it**.
4. Choose **UEFI** firmware. Leave Secure Boot enabled; Ubuntu and Fedora both boot cleanly under it.
5. Click **Customize Settings**, name the VM LAB-UBSRV01, and save it in your VM library folder.
6. Adjust settings before first boot:

   Processors & Memory: 2 cores, 2048 MB (4096 MB for a graphical desktop).

   Hard Disk: 40 GB, thin, single file.

   Network Adapter: **Share with my Mac**.
7. Power on and select **Try or Install Ubuntu Server** at the boot menu.
8. Complete the Subiquity installer: language, keyboard, default server install, DHCP networking, no proxy, default mirror, **Use an entire disk**, profile details, and **Install OpenSSH server**. Skip the snaps.
9. Reboot when prompted and remove the installation medium if asked. If the VM boots into the installer again, power it off and disconnect the CD/DVD device in settings.
10. Log in and install the guest tools:

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y open-vm-tools
sudo systemctl enable --now open-vm-tools
```

```
ip addr show
sudo reboot
```

11. From the Mac, confirm you can reach the guest over SSH using the address it reported:

```
ssh <username>@<guest-ip>
```

12. Shut down cleanly and take a snapshot named Clean Build.

17.3 Fedora Variation

```
# After installing Fedora Server 44 from the aarch64 or x86_64 DVD:
sudo dnf upgrade --refresh -y
sudo dnf install -y open-vm-tools
sudo systemctl enable --now vmtoolsd

sudo systemctl enable --now sshd
sudo firewall-cmd --add-service=ssh --permanent
sudo firewall-cmd --reload

getenforce
ip addr show
```

17.4 Verify

VERIFY

The guest boots to a login prompt without the ISO attached.

uname -m reports aarch64 on an Apple Silicon host, or x86_64 on an Intel Mac.

systemctl status open-vm-tools (or vmtoolsd on Fedora) is active.

SSH from the Mac to the guest succeeds.

A Clean Build snapshot exists.

17.5 Troubleshooting

Symptom	Likely cause	Fix
ISO will not boot at all on Apple Silicon	x86-64 image on an ARM64 host	Download the arm64 or aarch64 build
Console resolution fixed and tiny	open-vm-tools not installed	Install open-vm-tools, plus open-vm-tools-desktop for GUI guests, then reboot
No address from DHCP	Adapter not connected	Check the Network Adapter setting and confirm it is connected
SSH refused on Fedora	firewalld blocking or sshd disabled	sudo systemctl enable --now sshd, then open the service in firewalld
Installer cannot see the disk	Controller mismatch after manual edits	Recreate the VM with default controller settings rather than hand-editing the .vmx

Part 5: UTM (macOS)

UTM is a free, open-source virtual machine application for macOS, iOS, and visionOS, built on QEMU and Apple's Virtualization framework. It is Apache-licensed, it has no account requirement, and it does something none of the other three products in this guide can do: it can **emulate** a different processor architecture as well as virtualize the native one.

Virtualization versus Emulation, and Why UTM Has Both

	Virtualize	Emulate
Backend	Apple Virtualization framework, or QEMU with the Hypervisor framework	QEMU with dynamic translation
Guest architecture	Must match the host: ARM64 on Apple Silicon, x86-64 on Intel	Any of 30-plus architectures: x86-64, ARM32, MIPS, PowerPC, RISC-V, SPARC
Speed	Near native	**Roughly 5x to 20x slower**
Use it for	Everything you normally do	Running x86-64 on Apple Silicon when there is no ARM64 alternative

That emulation capability is genuinely useful and it is genuinely slow. It is the reason UTM is the only product in this guide that can run Windows Server 2025 on an Apple Silicon Mac at all. It is also the reason you should not plan a course around doing so.

> **WARNING**
> An emulated x86-64 Windows Server installation on Apple Silicon can take several hours where the same install takes fifteen minutes natively. Treat emulation as a capability you demonstrate once, not a platform you teach on.

UTM versus Fusion on a Mac

Consideration	UTM	VMware Fusion
Cost	Free from the website; $9.99 on the Mac App Store for automatic updates	Free, requires a Broadcom account
Licence	Apache 2.0, open source	Proprietary, free of charge
x86 emulation on Apple Silicon	**Yes**, via QEMU	No
Snapshots	Yes (QEMU backend)	Yes, with a full snapshot manager
Guest tools maturity	SPICE guest tools, good on Linux, more limited on Windows	VMware Tools, mature on both
Windows graphics	No 3D acceleration for Windows guests	Better, though still not a gaming platform
Best for	Free labs, unusual architectures,	Windows-heavy labs, parity with

Consideration	UTM	VMware Fusion
	lightweight Linux guests	Workstation on PCs

> **TIP**
> Install both. They coexist without conflict, they use different VM formats, and having both on hand means you can show students the difference between virtualization and emulation on the same machine in the same class.

18. Downloading UTM from mac.getutm.app

18.1 Explain

UTM is distributed three ways, and all three are the same software. The Mac App Store version costs $9.99 and the money funds development; it is functionally identical to the free build and adds automatic updates. The direct download is free and updates manually. Homebrew is free and updates through brew.

```
Official site:
https://mac.getutm.app

Source and release archive:
https://github.com/utmapp/UTM

Documentation:
https://docs.getutm.app

SPICE guest tools and support downloads:
https://mac.getutm.app/support/
```

18.2 Perform

Option A: Direct download (free)

1. Open https://mac.getutm.app in a browser.
2. Click **Download**. This retrieves UTM.dmg.
3. Note the version. As of this writing the current release is **UTM 4.7.5**, published in January 2026.

Option B: Mac App Store ($9.99)

1. Open the App Store and search for **UTM Virtual Machines**.
2. Purchase and install. You get automatic updates and you support the project.

Option C: Homebrew

1. From Terminal:

```
brew install --cask utm
```

Also download the guest support tools

1. From https://mac.getutm.app/support/, download the **SPICE guest tools ISO**. You will need it for Windows guests, and it is convenient to have on hand for Linux guests too.

18.3 Verify

> **VERIFY**
>
> You have UTM.dmg (direct download), or UTM appears in Applications (App Store or Homebrew).
>
> The version matches what the site advertises.
>
> If you plan to build a Windows guest, you also have the SPICE guest tools ISO.

19. Installing UTM on a MacBook

19.1 Explain

UTM is a sandboxed macOS application. Installation is a drag and drop, and there is no kernel extension to approve, which makes it the least invasive install in this guide. The permissions it needs are ordinary macOS privacy permissions.

19.2 Perform

1. Double-click UTM.dmg to mount it.
2. Drag **UTM.app** into the **Applications** folder shortcut in the mounted window.
3. Eject the disk image.
4. Open Applications and launch **UTM**. If macOS warns that it was downloaded from the internet, click **Open**.

   If Gatekeeper blocks it outright, go to System Settings > Privacy & Security, scroll to the security section, and click **Open Anyway**.
5. Grant permissions as UTM requests them, and check these under **System Settings > Privacy & Security**:

   Files and Folders, so UTM can read your ISO library and write VM bundles.

   Local Network, so bridged guests can reach other devices.

   Microphone and **Camera**, only if you intend to pass those devices into a guest. Otherwise leave them off.
6. Open **UTM > Settings** and set the default VM storage location to a folder outside iCloud Drive, Desktop, and Documents.

> **WARNING**
> Same rule as everywhere else in this guide. A VM bundle inside an iCloud-synced Desktop or Documents folder will be uploaded while it is running. Keep VMs in something like ~/VMs/ and confirm that folder is not syncing.

19.3 Verify

```
# Confirm the app is present and note the version:
ls -d /Applications/UTM.app
defaults read /Applications/UTM.app/Contents/Info.plist
CFBundleShortVersionString
```

> **VERIFY**
> UTM opens to an empty VM list with a **Create a New Virtual Machine** button.
>
> The version string matches what you downloaded.
>
> No Gatekeeper warning remains outstanding.

19.4 Troubleshooting

Symptom	Likely cause	Fix
"UTM cannot be opened because the developer cannot be verified"	Gatekeeper quarantine	System Settings > Privacy & Security > Open Anyway
UTM cannot see ISO files	Files and Folders permission not granted	Grant access under Privacy & Security, or move the ISO library somewhere UTM can read
App Store version behaves differently	It does not; the builds are identical	No action. The only difference is automatic updates.
Bridged networking shows no devices	Local Network permission denied	System Settings > Privacy & Security > Local Network, enable UTM

20. Creating a Windows Virtual Machine with UTM

20.1 Explain

UTM presents a choice on the first screen of the new VM wizard, and it is the most consequential click in Part 5.

- **Virtualize** runs the guest on the host CPU at near native speed. On Apple Silicon this means an ARM64 guest. On an Intel Mac it means an x86-64 guest.
- **Emulate** translates a foreign architecture. On Apple Silicon this is how you run x86-64 Windows, at a heavy speed cost.

Your Mac	Windows guest	Choose	Realistic install time
Intel Mac	Windows 11 x64 or Windows Server 2025 x64	Virtualize	20 to 40 minutes
Apple Silicon	Windows 11 ARM64	**Virtualize**	20 to 40 minutes
Apple Silicon	Windows Server 2025 x64	Emulate	**Several hours. Demonstration only.**

20.2 Procedure: Windows 11 ARM64 on Apple Silicon (recommended)

1. Obtain Windows 11 ARM64 media. UTM's own documentation walks through the current options at https://docs.getutm.app/guides/windows/, and Microsoft publishes ARM64 images from https://www.microsoft.com/software-download/.

   UTM historically included a helper that fetches and prepares the media for you. If your build offers it, use it.

2. In UTM click the **+** button, then **Create a New Virtual Machine**.
3. Select **Virtualize**.
4. Select **Windows**.
5. Under **Boot ISO Image**, click **Browse** and select your Windows 11 ARM64 ISO.
6. Leave **Install drivers and SPICE tools** ticked.

   This matters. It attaches the driver and tools image so you can install it after Windows is up. Without those drivers the guest has no proper display, no shared clipboard, and often no network.

7. Click **Continue**. Set **Memory** to **4096 MB** and leave CPU cores on the default so UTM manages allocation.
8. Set the storage size to **64 GB** or more, then click **Continue**.
9. Optionally choose a shared folder on the Mac, then click **Continue**.
10. Name the VM LAB-W11-ARM, review the summary, and click **Save**.
11. Select the VM and click the **play** button to start it.

12. Complete Windows Setup. If setup stalls on a network requirement, press **Shift+F10** at the language screen and run:

    ```
    OOBE\BYPASSNRO
    ```

    The VM restarts and offers "I don't have internet", then "Continue with limited setup".

13. Once the desktop appears, open File Explorer, find the mounted **SPICE guest tools** drive, and run the installer inside the guest. Reboot the guest afterward.

14. Apply Windows Updates, shut down cleanly, then create a snapshot from the VM settings.

[SCREENSHOT REQUIRED]
Capture: The UTM new virtual machine wizard on the Virtualize / Emulate selection screen
Show: Both options visible with Virtualize selected, on an Apple Silicon Mac
Caption: Figure 20-1: Choosing Virtualize rather than Emulate in UTM

20.3 Procedure: Emulated x86-64 Windows Server on Apple Silicon

Use this only when you specifically need to demonstrate x86-64 Windows Server on Apple hardware and no other host is available. Budget several hours and set expectations with the class first.

1. In UTM click **+**, then **Create a New Virtual Machine**.
2. Select **Emulate**.
3. Select **Windows**.
4. Browse to the Windows Server 2025 x64 evaluation ISO from Section 3.4.
5. Keep **Install drivers and SPICE tools** ticked.
6. Set memory to **4096 MB** and CPU cores to **4**. More cores helps emulation more than it helps virtualization.
7. Set storage to **80 GB** and continue.
8. Name the VM LAB-WS2025-EMULATED so nobody mistakes it for a normal build. Save.
9. Start the VM and complete the Windows Server installer. Expect every screen transition to be slow. This is normal and is not a fault.
10. After installation, install the SPICE guest tools from the mounted drive, then reboot.

> **NOTE**
> Set expectations with students explicitly. Emulation is a demonstration of what a processor architecture boundary actually costs. It is a memorable lesson precisely because it is so slow, and it makes the case for choosing the right host hardware better than any slide could.

20.4 Verify

> **VERIFY**

The guest boots to the Windows desktop.

System > About reports the expected architecture: ARM-based on a virtualized Apple Silicon guest, x64-based on an emulated one.

After installing the SPICE guest tools, the display resizes with the window and the clipboard works.

The guest has network connectivity.

20.5 Troubleshooting

Symptom	Likely cause	Fix
Windows Setup finds no disk	Storage driver not loaded	Ensure "Install drivers and SPICE tools" was ticked; at the disk screen use **Load driver** and point at the mounted tools image
No network in the guest	Driver not installed yet	Complete setup offline with OOBE\BYPASSNRO, then install the SPICE guest tools
Setup fails a network check	Windows 11 requires connectivity by default	Shift+F10 at the language screen, run OOBE\BYPASSNRO, choose limited setup
Boots to a UEFI shell	Boot order or a detached ISO	Type exit, use the Boot Manager, and select the CD device; confirm the ISO is still attached in VM settings
Everything is extremely slow	You selected Emulate rather than Virtualize	Delete and recreate the VM using Virtualize with ARM64 media
Graphics tear or applications render oddly	No 3D acceleration for Windows guests in UTM	Expected limitation. Use Fusion if graphics fidelity matters.

21. Creating a Linux Virtual Machine with UTM

21.1 Explain

This is where UTM is at its best. On Apple Silicon, an ARM64 Linux guest runs through Apple's Virtualization framework at near native speed, boots in seconds, and idles on very little memory. A Linux VM in UTM is the lightest lab environment available on a Mac.

UTM gives you a backend choice for Linux that it does not give you for Windows:

Backend	What it gives you	What it costs you
Apple Virtualization	Best performance, efficient memory use, Rosetta support for running x86-64 Linux binaries inside an ARM64 guest	Fewer device options, and **no snapshot support**
QEMU (default)	More virtual hardware options, broader guest support, **snapshots**	Slightly lower performance

> **TIP**
> For a teaching lab, use the QEMU backend. Losing snapshots to gain a few percent of performance is a bad trade in a classroom, where reverting a broken VM is worth more than any benchmark.

21.2 Prerequisites

- On Apple Silicon: ubuntu-26.04-live-server-arm64.iso or Fedora-Server-dvd-aarch64-44-*.iso.
- On an Intel Mac: the amd64 / x86_64 equivalents.
- At least 2 GB of free host RAM and 20 GB of free disk.

21.3 Perform

1. In UTM click **+**, then **Create a New Virtual Machine**.
2. Select **Virtualize**.
3. Select **Linux**.
4. Leave **Use Apple Virtualization** unticked to keep the QEMU backend and retain snapshots.
5. Under **Boot ISO Image**, click **Browse** and select your ARM64 (or x86_64) Linux ISO.
6. Click **Continue**. Set memory to **2048 MB** and leave CPU cores at the default.
   Use 4096 MB if you are installing a desktop environment.
7. Set storage to **40 GB** and click **Continue**.
8. Optionally set a shared directory on the Mac, then click **Continue**.
9. Name the VM LAB-UBSRV01, review the summary, and click **Save**.
10. Start the VM. If you land at a UEFI shell, type exit, choose **Boot Manager**, and select the CD device.

11. Complete the Ubuntu Server installer: language, keyboard, default server install, DHCP networking, no proxy, default mirror, **Use an entire disk**, profile details, and **Install OpenSSH server**.
12. Reboot when prompted. If the VM boots back into the installer, shut it down, open the VM settings, remove the ISO from the CD/DVD drive, and start it again.
13. Log in and finish the guest setup:

```
sudo apt update && sudo apt upgrade -y

# SPICE agent gives clipboard sharing and dynamic resolution:
sudo apt install -y spice-vdagent qemu-guest-agent
sudo systemctl enable --now qemu-guest-agent

uname -m        # aarch64 on Apple Silicon
ip addr show
sudo reboot
```

14. From the Mac, confirm SSH access to the guest, then shut it down cleanly and take a snapshot from the VM settings.

21.4 Fedora Variation

```
# Fedora Server 44, aarch64 or x86_64:
sudo dnf upgrade --refresh -y
sudo dnf install -y spice-vdagent qemu-guest-agent
sudo systemctl enable --now qemu-guest-agent

sudo systemctl enable --now sshd
sudo firewall-cmd --add-service=ssh --permanent
sudo firewall-cmd --reload

getenforce
uname -m
```

21.5 A Note on the UTM Gallery

UTM publishes a gallery of prebuilt virtual machines at https://mac.getutm.app/gallery/, which can be imported with a click. They are convenient for a quick demonstration.

> **WARNING**
> Prebuilt images are somebody else's build. You do not know what is installed, what credentials exist, or what the image phones home to. For a security course, the correct default is to build from a verified vendor ISO. Use the gallery to show students how easy import is, then explain precisely why you would not use it for anything that matters.

21.6 Verify

> **VERIFY**
> The guest boots to a login prompt with no ISO attached.

uname -m reports aarch64 on Apple Silicon or x86_64 on Intel.

The console resizes with the window and the clipboard works, which confirms spice-vdagent.

SSH from the Mac to the guest succeeds.

A snapshot exists (QEMU backend only).

21.7 Troubleshooting

Symptom	Likely cause	Fix
Boots to a UEFI shell	Boot order	Type exit, choose Boot Manager, select the CD device
Blank screen or a boot loop on ARM64	Display device incompatibility	In VM settings > Display, switch the emulated display card to virtio-ramfb (or virtio-gpu-pci) and restart
No snapshot option anywhere	Apple Virtualization backend selected	Recreate the VM with the QEMU backend; Apple Virtualization does not support snapshots
Clipboard and resizing do not work	spice-vdagent missing	Install spice-vdagent in the guest and reboot
Guest is unexpectedly slow	Emulate selected instead of Virtualize, or wrong-architecture media	Confirm uname -m matches the host architecture; rebuild with matching media
Shared directory not visible in the guest	VirtFS or SPICE WebDAV not mounted	Follow the sharing guide at https://docs.getutm.app; the mechanism differs per backend

Part 6: Reference and Lab Operations

22. Host Virtualization Conflicts: The Master Checklist

Roughly half of all "my virtual machine will not start" reports trace back to something in this section. Work it top to bottom before troubleshooting anything else.

22.1 Firmware

Reboot into UEFI/BIOS setup and confirm the virtualization extensions are enabled. The setting name varies by vendor:

Vendor	Typical setting name	Where to look
Intel systems	Intel Virtualization Technology, Intel VT-x, VT-d	Advanced, CPU Configuration, or Security
AMD systems	SVM Mode, AMD-V, SVM Support	Advanced, CPU Configuration, or M.I.T.
Dell	Virtualization Support > Virtualization	System Setup
HP	Virtualization Technology (VTx)	Security or System Configuration
Lenovo / ThinkPad	Intel Virtualization Technology	Security > Virtualization

Confirming from the operating system

```
# Windows PowerShell
Get-ComputerInfo -Property "HyperVRequirement*"

# Linux: a non-zero count means the extensions are present
grep -Ec "vmx|svm" /proc/cpuinfo
lscpu | grep -i virtual

# macOS: 1 means supported
sysctl -n kern.hv_support
```

22.2 Windows Features That Claim the Hypervisor

Any one of these turns on the Windows hypervisor, and other Type 2 products then run on top of it rather than on bare hardware.

Feature	How to check	How to disable
Hyper-V platform	Task Manager > Performance > CPU shows "Virtual machine: Yes"	Disable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V-All
Hypervisor launch at boot	bcdedit /enum {current}	bcdedit /set hypervisorlaunchtype off
WSL2	wsl --status	Disable **Virtual Machine Platform**, or convert distributions to WSL1

Feature	How to check	How to disable
Windows Sandbox	Optional Features list	Disable-WindowsOptionalFeature -Online -FeatureName Containers-DisposableClientVM
Memory Integrity	Windows Security > Device security > Core isolation	Toggle off in the same place, then reboot
Credential Guard	Get-CimInstance Win32_DeviceGuard	Often enforced by Group Policy. Ask your administrator.

> **WARNING**
> These are security features. Disabling Memory Integrity and Credential Guard measurably weakens a Windows host against credential theft. That is a defensible trade on a dedicated, isolated lab machine. It is not a defensible trade on the laptop you use for work email. Where possible, keep lab and production hosts separate.

22.3 Two Hypervisors at Once

You can install VMware Workstation and VirtualBox on the same host. You should not run virtual machines in both at the same time. Both want exclusive use of the CPU virtualization extensions, and the result ranges from mysterious slowness to a host bug check. Shut down all VMs in one product before starting VMs in the other.

22.4 Endpoint Protection

Real-time antivirus scanning of virtual disk files is a common and invisible cause of terrible VM performance. Every guest write becomes a host file write, and the scanner inspects it.

- Exclude your VM library folder from real-time scanning.
- Exclude your ISO library folder.
- Do **not** exclude anything inside the guest. Guest-side protection is a separate question and usually should stay on.

> **WARNING**
> If your lab handles malware samples, do not exclude anything without a considered decision and, in an organizational setting, without approval. Exclusions in a malware analysis lab need to be paired with genuine network isolation, discussed next.

23. Virtual Networking Modes and Lab Isolation

Every product in this guide offers the same three fundamental modes under different names. Choosing correctly is a security decision, not a convenience decision.

Mode	VMware name	VirtualBox name	UTM name	What it does
NAT	NAT / Share with my Mac	NAT	Shared Network	Guest reaches the internet through the host. Other machines cannot reach the guest without port forwarding.
Bridged	Bridged	Bridged Adapter	Bridged	Guest appears as a peer on the physical LAN with its own address.
Isolated	Host-only / LAN segment	Host-only / Internal Network	Host Only / Emulated VLAN	Guest talks only to the host, or only to other guests. No path to the physical network.

23.1 Choosing a Mode

Lab scenario	Use	Why
Installing and patching a fresh guest	NAT	Outbound access for updates, no exposure on the LAN
Active Directory lab with several guests	Host-only or internal network, plus one NAT adapter on a single gateway VM	Guests see each other, the domain is not exposed to the LAN
A guest that must be reachable from other physical machines	Bridged	The guest gets a real LAN address
Malware analysis or unknown binaries	**Isolated internal network only, with no NAT adapter**	A detonation VM on NAT or bridged can reach the internet and your LAN. This is how lab incidents become real incidents.
Packet capture exercises between guests	Internal network	Traffic stays inside the virtual switch and is easy to observe

> **WARNING**
> The single most serious mistake a student can make in a virtualization lab is running untrusted code in a guest that is bridged to the campus or home network. NAT is not isolation either; a NAT guest still reaches the internet and can still reach hosts on the physical LAN in many configurations. If the exercise involves anything untrusted, use an isolated internal network with no route out, verify it, and verify it again by trying to ping something and failing.

23.2 Verifying Isolation

Teach students to prove isolation rather than assume it. From inside the guest:

```
# All three of these should FAIL on a properly isolated guest
ping -c 3 8.8.8.8
```

```
ping -c 3 <your host physical IP>
curl -m 5 https://example.com

# And this should succeed, to other guests on the same isolated segment
ping -c 3 <other guest IP>
```

> **TIP**
> Make "prove the isolation" a graded step in every lab that touches untrusted code. A screenshot of three failed pings is worth more than a paragraph of assurance.

24. Lab Hygiene

24.1 Naming

Adopt a convention on day one and require it. Something like <ROLE>-<NUMBER>-<OS> produces LAB-DC01-WS2025, LAB-CLI01-W11, LAB-SRV02-UBSRV2604. It reads correctly in a VM list, in a snapshot tree, and in a student's report six weeks later.

24.2 The Snapshot Discipline

Snapshot name	When to take it	Keep for
Clean Build	Immediately after OS install, guest tools, and updates, with the guest powered off	The life of the VM
Pre-<exercise>	Before a lab that changes system state	Until the exercise is graded
Working-<date>	At a checkpoint inside a long exercise	Delete when the exercise is done

- Take snapshots with the guest **powered off** when you can. A powered-off snapshot is smaller, faster, and always consistent.
- Keep the chain short. Two or three per VM. Delete working snapshots when the exercise ends.
- **A snapshot is not a backup.** It lives in the same folder, on the same disk, in the same VM. Lose the folder and you lose every snapshot with it.

24.3 Backups

The real backup of a virtual machine is a copy of its folder on separate storage, taken while the VM is powered off. Everything else is a convenience.

```
# Windows: robocopy mirrors the VM folder to an external drive
robocopy "D:\VMs\LAB-DC01-WS2025" "E:\Backup\LAB-DC01-WS2025" /MIR /R:1 /W:1

# Linux and macOS
rsync -aH --delete ~/VMs/LAB-UBSRV01/ /Volumes/Backup/LAB-UBSRV01/
```

```
# Or export a portable appliance (works in VirtualBox, imports into VMware)
VBoxManage export LAB-UBSRV01 -o LAB-UBSRV01.ova
```

24.4 Reclaiming Disk Space

Thin disks grow but do not shrink on their own. When the guest deletes files, the host file stays the same size. Compact it periodically.

```
# VMware: zero free space in the guest first, then compact from the host
#   Linux guest:   sudo dd if=/dev/zero of=/zero bs=1M; sudo rm -f /zero
#   Windows guest: use the VMware Tools shrink option, or sdelete -z
vmware-vdiskmanager -k "LAB-UBSRV01.vmdk"

# VirtualBox
VBoxManage modifymedium disk "LAB-UBSRV01.vdi" --compact
```

24.5 Decommissioning

Deleting a VM from the interface usually deletes the files. Deleting the folder by hand usually leaves an orphaned registration. Do it from the product, then confirm the folder is gone, then confirm your backup copy is either kept deliberately or removed deliberately.

> **WARNING**
> If a lab VM ever held real data, real credentials, or a real capture, treat its virtual disk like any other storage media at end of life. Deleting the file is not erasure on an SSD with wear levelling. Encrypt the VM from the start and destroy the key, or use full disk encryption on the host so the question never arises.

25. Choosing a Hypervisor

	VMware Workstation	VirtualBox	VMware Fusion	UTM
Host platforms	Windows, Linux	Windows, Linux, macOS, Solaris	macOS	macOS, iOS
Cost	Free	Free (base)	Free	Free, or $9.99 App Store
Account required	**Yes**, Broadcom	No	**Yes**, Broadcom	No
Licence	Proprietary	GPLv3 base, proprietary extension pack	Proprietary	Apache 2.0
Apple Silicon	No	Yes (ARM64 guests)	Yes (ARM64 guests)	Yes
x86 emulation on ARM	No	No	No	**Yes**

	VMware Workstation	VirtualBox	VMware Fusion	UTM
Snapshots	Yes, full manager	Yes, with branching	Yes, full manager	Yes (QEMU backend only)
Fedora module handling	Manual rebuild per kernel	**DKMS, automatic**	Not applicable	Not applicable
Best fit	Windows and Linux PC labs	Mixed-hardware classes	Mac labs needing Windows	Free Mac labs, odd architectures

A short recommendation

- **Class where everyone has a Windows or Linux PC:** VMware Workstation, with VirtualBox as the fallback for anyone who cannot disable Hyper-V.
- **Class with a mix of PCs and Macs:** VirtualBox everywhere, for one set of instructions and one VM format.
- **Mac-heavy class needing Windows guests:** VMware Fusion.
- **Mac class on a zero-budget footing, or teaching architecture:** UTM.

Appendices

Appendix A: Command Quick Reference

VMware Workstation and Fusion

```
vmrun -T ws list                          # running VMs (Workstation)
vmrun -T fusion list                      # running VMs (Fusion)
vmrun -T ws start "C:\VMs\LAB.vmx" nogui
vmrun -T ws stop  "C:\VMs\LAB.vmx" soft
vmrun -T ws snapshot "C:\VMs\LAB.vmx" "Clean Build"
vmrun -T ws revertToSnapshot "C:\VMs\LAB.vmx" "Clean Build"
vmware-vdiskmanager -k disk.vmdk          # compact a thin disk

# Linux host maintenance
sudo vmware-modconfig --console --install-all
lsmod | grep -E "vmmon|vmnet"
```

VirtualBox

```
VBoxManage --version
VBoxManage list vms
VBoxManage list runningvms
VBoxManage list extpacks
VBoxManage list hostinfo
VBoxManage startvm "LAB-UBSRV01" --type headless
VBoxManage controlvm "LAB-UBSRV01" acpipowerbutton
VBoxManage snapshot "LAB-UBSRV01" take "Clean Build"
VBoxManage snapshot "LAB-UBSRV01" restore "Clean Build"
VBoxManage modifymedium disk "LAB-UBSRV01.vdi" --compact
VBoxManage export "LAB-UBSRV01" -o LAB-UBSRV01.ova

# Linux host maintenance
sudo /sbin/vboxconfig
lsmod | grep vboxdrv
```

Guest-side checks

```
# Linux
uname -m                                  # architecture
ip addr show                              # addresses
systemctl status open-vm-tools            # VMware guest tools
systemctl status vmtoolsd                 # VMware tools on Fedora
lsmod | grep vboxguest                    # VirtualBox additions
systemctl status qemu-guest-agent         # UTM / QEMU
getenforce                                # SELinux state (Fedora)
sudo firewall-cmd --list-all              # firewalld (Fedora)

# Windows (PowerShell in the guest)
Get-ComputerInfo -Property OsName,OsArchitecture,CsSystemType
Get-Service VMTools,VBoxService -ErrorAction SilentlyContinue
Test-NetConnection www.microsoft.com -Port 443
```

Checksum verification

```
Get-FileHash -Algorithm SHA256 .\image.iso     # Windows
sha256sum image.iso                            # Linux
sha256sum -c SHA256SUMS --ignore-missing       # Linux, whole file
shasum -a 256 image.iso                        # macOS
```

Appendix B: Download Sources at a Glance

What	Where	Account needed
VMware Workstation Pro	Broadcom Support Portal, VMware Cloud Foundation division, My Downloads	**Yes**, free Basic account with completed profile
VMware Fusion Pro	Same portal, same division	**Yes**
Oracle VirtualBox	https://www.virtualbox.org/wiki/Downloads	No
VirtualBox Extension Pack	Same page. **Free for personal and educational use only.**	No
UTM	https://mac.getutm.app, Mac App Store, or brew install --cask utm	No
Windows Server evaluation	https://www.microsoft.com/en-us/evalcenter	**Yes**, Microsoft account
Windows 11 (including ARM64)	https://www.microsoft.com/software-download/	Sometimes
Ubuntu Server	https://ubuntu.com/download/server, ARM64 at https://cdimage.ubuntu.com/releases/	No
Fedora Server	https://fedoraproject.org/server/download	No

Appendix C: Build Checklists

C.1 Before You Start (host readiness)

- Virtualization extensions enabled in firmware and confirmed from the OS
- Conflicting Windows hypervisor features disabled, or a decision recorded to accept the performance cost
- At least 16 GB RAM, and 250 GB free on an SSD
- VM library folder created **outside** any cloud sync folder
- ISO library folder created outside any cloud sync folder
- Antivirus exclusions applied to the VM and ISO folders
- Host fully updated and rebooted

C.2 Per Virtual Machine

- ISO downloaded from the vendor and **checksum verified**
- Architecture matches the host (x86-64 or ARM64)
- Named to convention
- Stored in the VM library folder
- Firmware set to UEFI; Secure Boot and TPM set deliberately

- vCPU set to 2 unless measured otherwise
- Memory set from the sizing table, not from the wizard maximum
- Thin virtual disk, sized generously, single file
- Network mode chosen deliberately and, for untrusted work, isolation proven
- Unattended or Easy Install declined
- Guest OS installed and updated
- Guest tools installed and verified
- Credentials recorded in your password manager
- Clean Build snapshot taken with the guest powered off
- Folder copied to separate storage as a real backup

C.3 Weekly Lab Maintenance

- Host disk free space above 15 percent
- Snapshot chains trimmed to two or three per VM
- Thin disks compacted if they have grown a lot
- Host and guest updates applied
- On Linux hosts, kernel modules rebuilt after any kernel update
- Backups verified by actually restoring one, not by looking at a folder listing

Appendix D: Student Lab Exercises

These map to the sections above and can be assigned individually or as a sequence. Each is written so the deliverable proves the student did the work rather than copied the outcome.

Lab 1: Host Capability Assessment

Objective. Determine whether a given host can support a virtualization lab, and document the reasoning.

Deliverable. A one-page report stating: CPU model and physical core count, total RAM, host reserve applied and why, resulting usable VM memory, storage type and free space, virtualization extension state with the command output that proves it, and any hypervisor conflicts found. Conclude with the maximum number of concurrent virtual machines this host can support and the sizing you would give each one.

Assessment. Full marks require the command output, not just the claim. A student who writes "virtualization is enabled" without evidence has not done the exercise.

Lab 2: Media Acquisition and Verification

Objective. Obtain and verify installation media from three vendors.

Deliverable. For a Windows Server evaluation ISO, an Ubuntu Server ISO, and a Fedora Server ISO: the source URL, the file size, the published SHA256, the computed SHA256, and a statement of whether they match. Include one paragraph on what you would do if they did not match, and why redownloading is not automatically the right answer.

Lab 3: First Build

Objective. Build one Linux server virtual machine from scratch in the hypervisor assigned to you.

Deliverable. Screenshots of the VM settings before first boot, the installer at the disk partitioning step, the guest reporting its IP address, uname -m output, guest tools running, and the snapshot manager showing Clean Build. Include the resource sizing you chose and one sentence justifying each number.

Lab 4: Windows Server Build and Resource Observation

Objective. Build a Windows Server VM and observe host resource impact.

Deliverable. Host memory and CPU utilization recorded at three points: host idle with no VMs, VM powered on and idle, VM under load during Windows Update. Include a short analysis of what changed and why, referencing the concepts in Section 1.

Lab 5: Isolation Proof

Objective. Configure and **prove** network isolation on a guest.

Deliverable. Configure a guest on an isolated internal network. Provide evidence that it cannot reach the internet, cannot reach the host's physical address, and can reach a second guest on the same segment. Then switch the same guest to NAT and repeat the tests, documenting exactly what changed. Conclude with one paragraph on why NAT is not isolation.

Lab 6: Snapshot and Recovery Drill

Objective. Demonstrate that recovery works, rather than assuming it does.

Deliverable. From a Clean Build snapshot, deliberately break the guest (rename a critical service, corrupt a config file, remove a package the system needs). Document the failure, then revert and document the recovery. Record how long the revert took. Then answer: if the host SSD failed right now, what would you have lost, and what would you do about it?

Lab 7: Cross-Platform Comparison

Objective. Build the same Linux guest in two different hypervisors and compare.

Deliverable. A comparison table covering install time, steps required, guest tools procedure, snapshot behavior, network mode names, and any difficulty encountered. Conclude with a recommendation for a specific scenario you define, and defend it.

Suggested Grading Rubric

Criterion	Weight	What full marks look like
Technical correctness	35%	The build works, the settings are appropriate, and the commands are right
Evidence quality	25%	Screenshots and command output actually prove the claims made
Reasoning	25%	Sizing and configuration choices are justified against the concepts in Part 1, not copied
Documentation	15%	A peer could reproduce the build from the submission alone

Appendix E: Glossary

Term	Definition
AMD-V	AMD's hardware virtualization extensions. The AMD equivalent of Intel VT-x. Sometimes labelled SVM in firmware.
Bridged networking	A virtual NIC connected directly to the physical network, so the guest gets its own address on the LAN.
DKMS	Dynamic Kernel Module Support. Rebuilds out-of-tree kernel modules automatically when the kernel changes. VirtualBox uses it; VMware Workstation does not.
Emulation	Translating instructions from one processor architecture to another in software. Correct but slow, typically 5x to 20x slower than virtualization.
EPT / NPT / RVI	Hardware second-level address translation. Lets the CPU translate guest-physical to host-physical addresses without software shadow page tables.
Extension Pack	The proprietary add-on for VirtualBox providing USB 2.0/3.0, RDP, disk encryption, and PXE. Free for personal and educational use only.
Guest	The operating system running inside a virtual machine.
Guest Additions	VirtualBox's in-guest driver and integration package.
Host	The physical machine and its operating system, running the hypervisor.
Host-only network	A virtual network connecting guests and the host, with no route to the physical network.
Hypervisor	The software layer that creates and runs virtual machines. Type 1 runs on bare metal; Type 2 runs as an application on a host OS.
ISO image	A byte-for-byte image of an optical disc in a single file, named for the ISO 9660 filesystem standard.
NAT networking	The guest reaches outside through the host's address. Inbound connections require explicit port forwarding.
open-vm-tools	The open-source VMware guest tools package, distributed by Linux distributions and preferred over the VMware-supplied tools for Linux guests.
Oversubscription	Assigning more virtual resources across all VMs than the host physically has. Normal for CPU, dangerous for memory.
Paravirtualized driver	A guest driver written to talk to the hypervisor directly rather than pretending to be physical hardware. Faster than emulated devices.
Rosetta	Apple's x86-64 translation layer. UTM and Fusion can expose it to Linux guests on Apple Silicon so x86-64 Linux binaries run inside an ARM64 guest.
Secure Boot	A UEFI feature that only allows signed boot components. Applies both to the host, where it affects module loading, and to guests.
Snapshot	A saved point-in-time state of a virtual machine. Subsequent writes go to a delta file. **Not a backup.**

Term	Definition
SPICE	A remote display protocol used by QEMU and UTM. spice-vdagent in the guest provides clipboard sharing and dynamic resolution.
Thin provisioning	A virtual disk file that grows as the guest writes, rather than occupying its declared size immediately.
TPM	Trusted Platform Module. A virtual TPM can be added to a guest, and Windows 11 requires one.
UEFI	The firmware interface that replaced legacy BIOS. Modern guests expect it.
vCPU	A virtual processor presented to a guest. A schedulable thread, not a dedicated physical core.
VMware Tools	VMware's in-guest driver and integration package. Required for sensible display, clipboard, and time handling.
VT-x	Intel's hardware virtualization extensions.
Virtualization	Running a guest natively on the host CPU under hypervisor supervision. Requires matching architectures.

Appendix F: Version Reference

Software versions current as of publication. Verify against the vendor before teaching from this guide, because all four products move.

Product or image	Version at publication	Released
VMware Workstation Pro	26H1	May 2026
VMware Fusion Pro	26H1	May 2026
Oracle VirtualBox	7.2.16	August 2026
UTM	4.7.5	January 2026
Windows Server evaluation	2025 (180-day evaluation)	GA November 2024
Ubuntu Server LTS	26.04 LTS, Resolute Raccoon	April 2026
Fedora Linux	44	April 2026

NOTE

Version drift is the main maintenance burden on a document like this. The procedures change far more slowly than the version numbers do. When you revise, check three things first: whether the Broadcom download path has moved again, whether VirtualBox has changed its Extension Pack licensing, and whether Microsoft has published an ARM64 Windows Server evaluation image. That last one would meaningfully change Part 4 and Part 5.